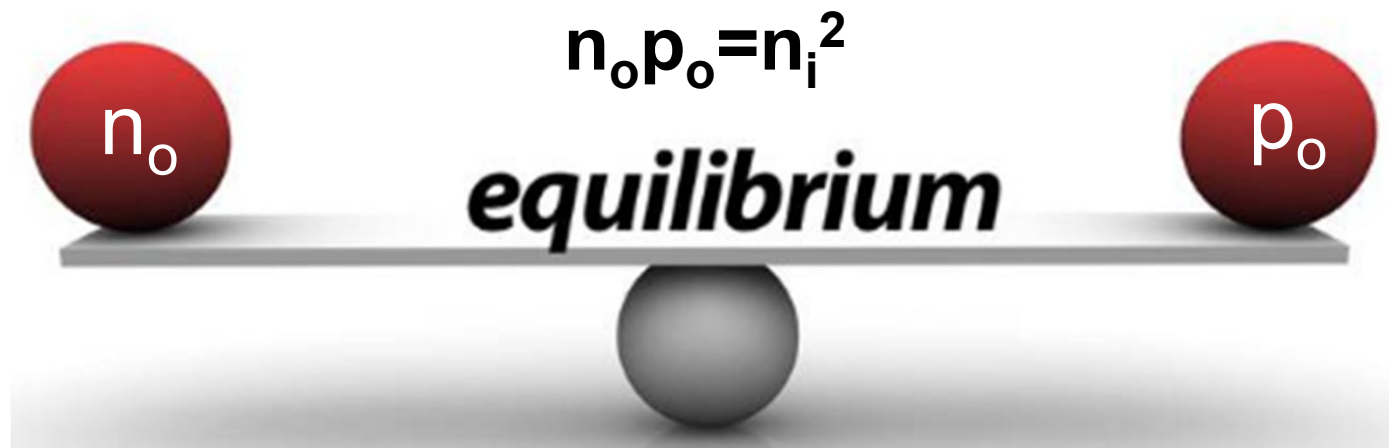


L4: The Semiconductor in Equilibrium



No external forces such as voltages, electrical fields, magnetic fields, or temperature gradients are acting on the semiconductor

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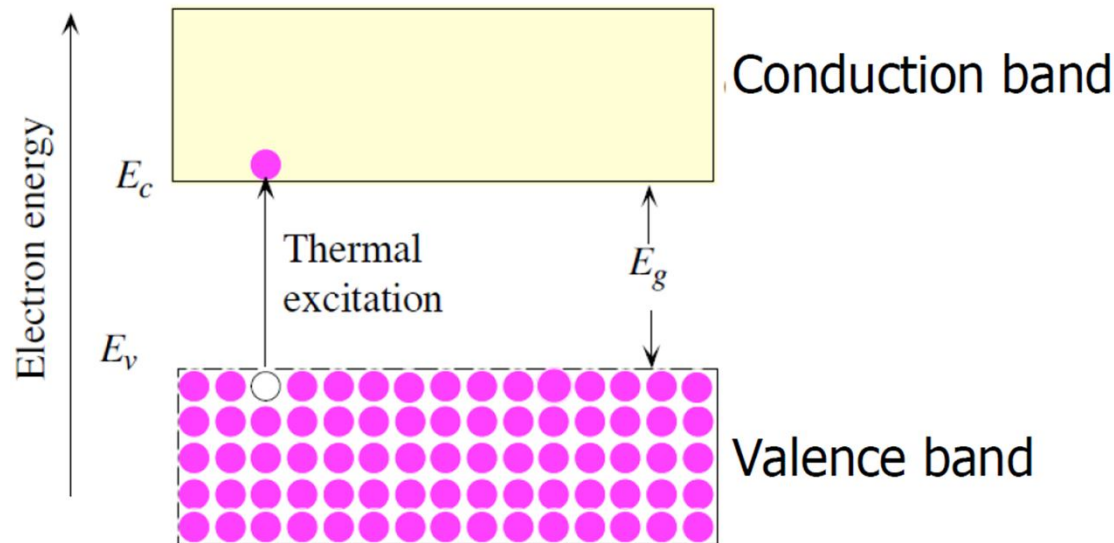
Topics

- Thermal-equilibrium concentration of electron and holes
- Intrinsic carrier concentration
- Intrinsic Fermi-level position
- Dopant atoms and energy levels
- Extrinsic carrier concentration and temperature dependence
- Ionization energy of dopant atoms in silicon
- Fermi level in extrinsic semiconductor
- Degenerate semiconductors
- Fermi level in two systems in contact with each other and at thermal equilibrium.

4.1 | CHARGE CARRIERS IN SEMICONDUCTORS

Why we want to know the number of carriers?

Current is determined by flow rate and density of charge carriers.



➤ Particles that can freely move and contribute to the current flow (conduction)
carrier

1. Electron in conduction band
2. Hole in valence band

$$J = nqv = nquE$$

- In this chapter, we will focus on n : the carrier density
- In next chapter, we will study the transport mechanism of carrier

How to count number of carriers?

Assumption; Pauli exclusion principle

If we know

1. No. of energy states



Density of states (DOS)

2. Occupied energy states



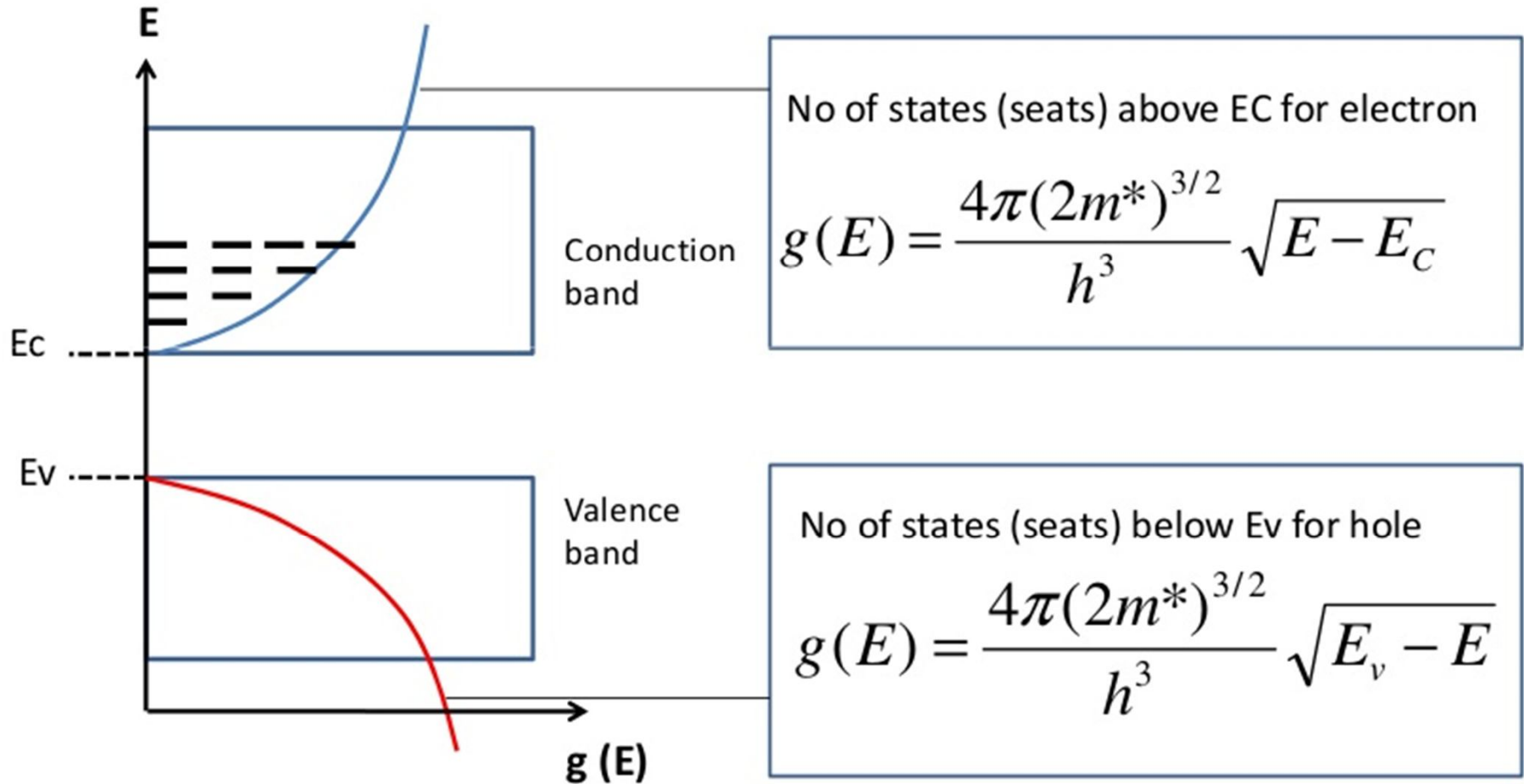
The probability that energy states is occupied

“Fermi-Dirac distribution function”



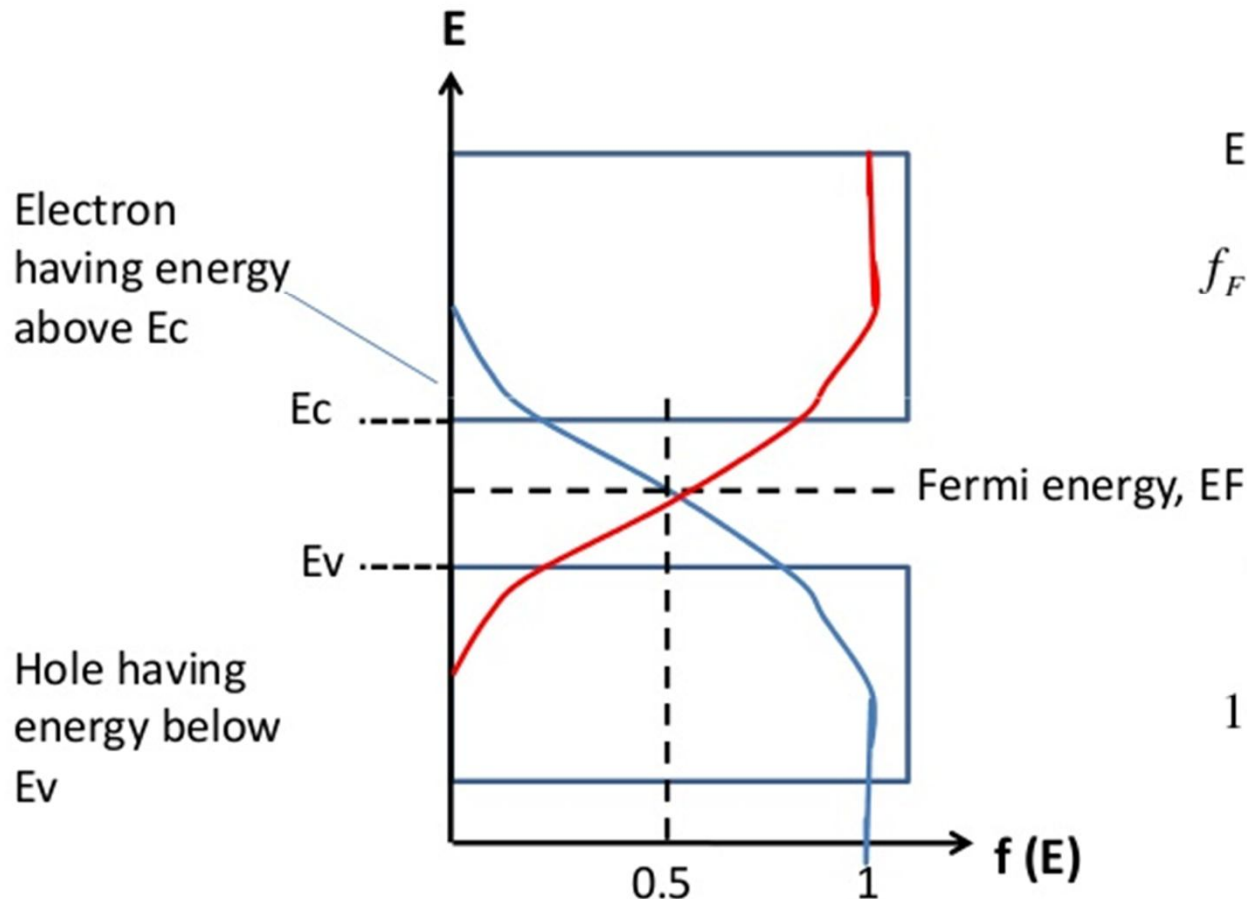
$$n = \text{DOS} \times \text{“Fermi-Dirac distribution function”}$$

Density of State



Fermi-Dirac distribution

Probability of electron having certain energy



Electron (blue line)

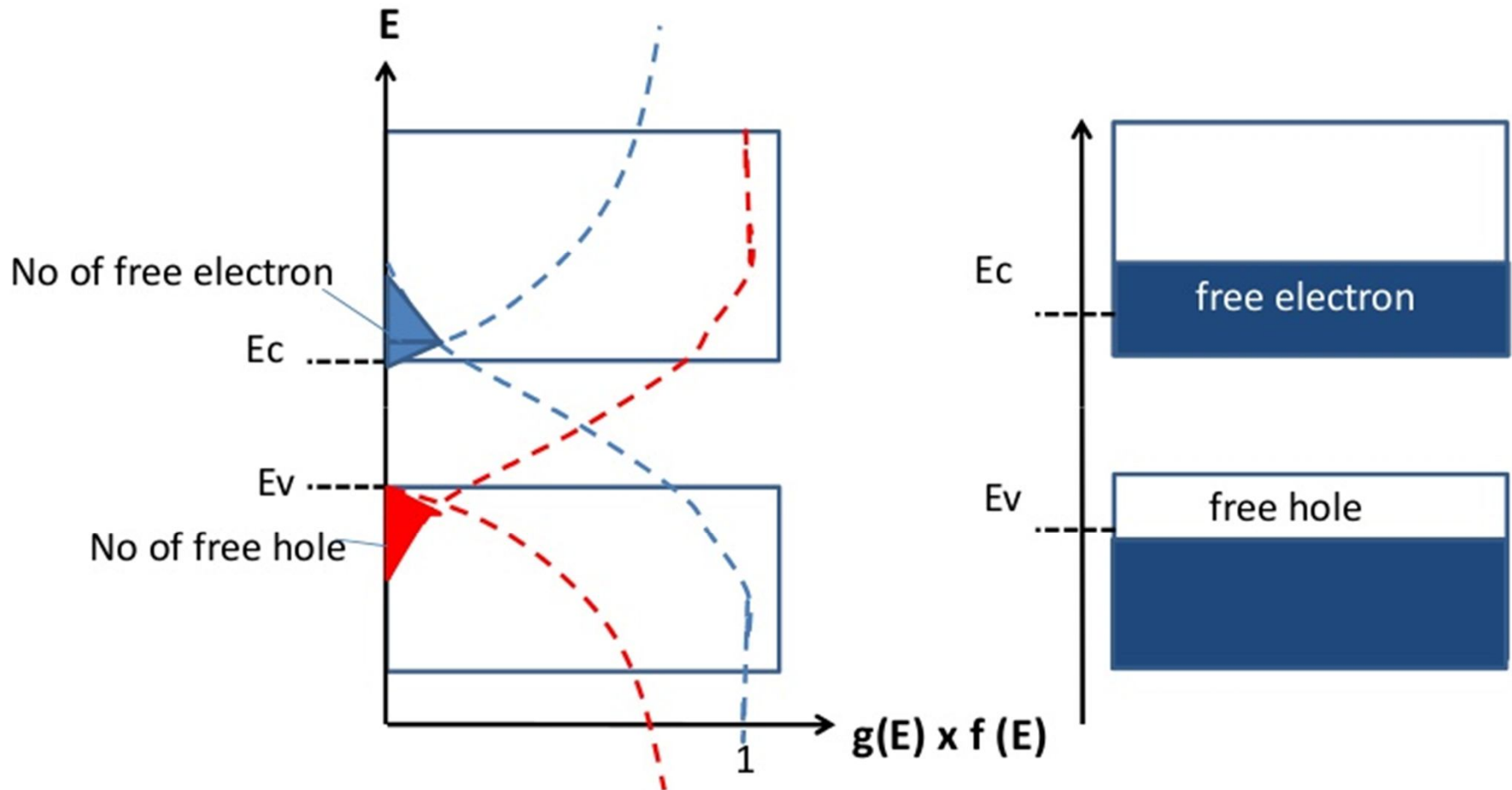
$$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

hole (red line)

$$1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

E_F ; the energy below which all states are filled with electron and above which all states are empty at 0K

Equilibrium Distribution of Electrons and Holes



Thermal-Equilibrium Electron Concentration

$$n_0 = \int g_c(E) f_F(E) dE$$

$$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

Condition:
 $E - E_F \gg kT$

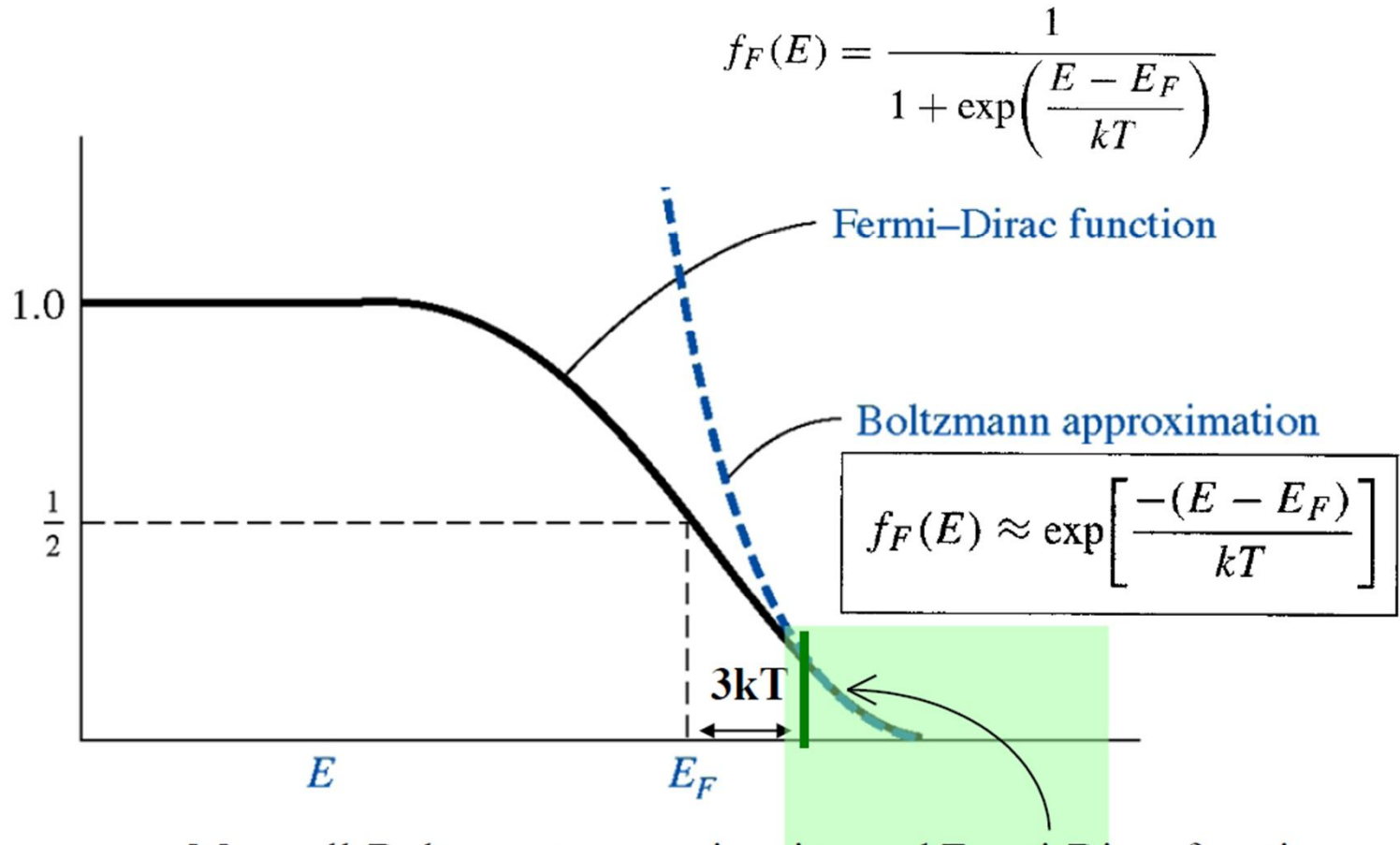


$$f_F(E) \approx \exp\left[\frac{-(E - E_F)}{kT}\right]$$

Boltzman approximation

Thermal-Equilibrium Electron Concentration

Comparison of Fermi-Dirac probability function and Maxwell-Boltzmann approximation



Maxwell-Boltzmann approximation and Fermi-Dirac function are within 5% of each other when $E - E_F \geq 3kT$

Thermal-Equilibrium Electron Concentration

$$n_0 = \int_{E_c}^{\infty} \frac{4\pi (2m_n^*)^{3/2}}{h^3} \sqrt{E - E_c} \exp \left[\frac{-(E - E_F)}{kT} \right] dE$$

$$\eta = \frac{E - E_c}{kT}$$

$$n_0 = \frac{4\pi (2m_n^* kT)^{3/2}}{h^3} \exp \left[\frac{-(E_c - E_F)}{kT} \right] \int_0^{\infty} \eta^{1/2} \exp(-\eta) d\eta$$

Gamma function:

$$\frac{1}{2} \sqrt{\pi}$$

Thermal-Equilibrium Electron Concentration

$$n_0 = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2} \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$

$$N_c = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2}$$

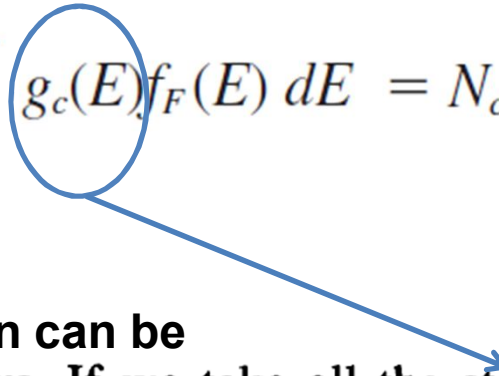
Dependent on temperature

N_c = effective density of states function in the conduction band

$$n_0 = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$

The equation is valid for **both intrinsic and extrinsic semiconductors**

Thermal-Equilibrium Electron Concentration

$$n_0 = \int g_c(E) f_F(E) dE = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$


The above equation can be interpreted as follows. If we take all the states in the conduction band and replace them with an effective concentration N_c (number of states per unit volume) at E_c and then multiply this simply by the Boltzmann probability function, $f(E_c) = \exp[-(E_c - E_F)/kT]$, we obtain the concentration of electrons at E_c , that is, in the conduction band. N_c is thus an effective density of states at the CB band edge.

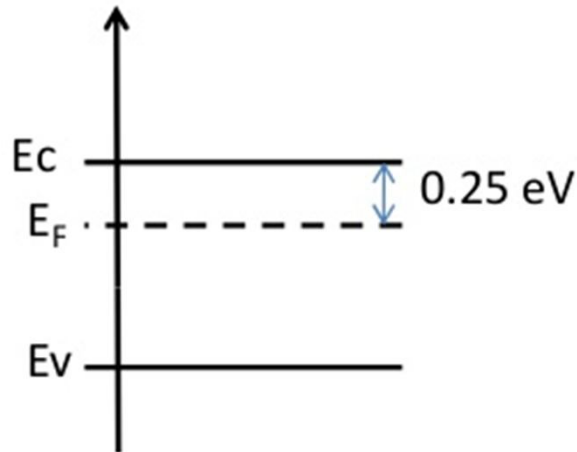
导带电子浓度可理解为：把导带中所有的量子态都集中在导带底 E_c ，而它的有效状态密度为 N_c ，则导带中的电子浓度就是服从波尔兹曼分布的 N_c 个状态中有电子占据的量子态数。

Thermal-Equilibrium Electron Concentration

Ex. 1

Calculate the thermal equilibrium electron concentration in Si at $T = 300\text{K}$.

Assume that Fermi energy is 0.25 eV below the conduction band. The value of N_C for Si at $T = 300\text{K}$ is $2.8 \times 10^{19} \text{ cm}^{-3}$.



$$\begin{aligned} n_o &= N_C \exp\left[\frac{-(E_C - E_F)}{kT}\right] \\ &= 2.8 \times 10^{19} \cdot \exp\left[\frac{-(E_C - E_C + 0.25)}{0.0259}\right] \\ &= 1.8 \times 10^{15} \text{ cm}^{-3} \end{aligned}$$

Thermal-Equilibrium Hole Concentration

$$p_0 = \int g_v(E)[1 - f_F(E)] dE$$

$$1 - f_F(E) = \frac{1}{1 + \exp\left(\frac{E_F - E}{kT}\right)}$$

$$1 - f_F(E) = \frac{1}{1 + \exp\left(\frac{E_F - E}{kT}\right)} \approx \exp\left[\frac{-(E_F - E)}{kT}\right]$$

Boltzman approximation

$$p_0 = \int_{-\infty}^{E_v} \frac{4\pi(2m_p^*)^{3/2}}{h^3} \sqrt{E_v - E} \exp\left[\frac{-(E_F - E)}{kT}\right] dE$$

Thermal-Equilibrium Hole Concentration

$$p_0 = 2 \left(\frac{2\pi m_p^* kT}{h^2} \right)^{3/2} \exp \left[\frac{-(E_F - E_v)}{kT} \right]$$

$$N_v = 2 \left(\frac{2\pi m_p^* kT}{h^2} \right)^{3/2}$$

N_v = effective density of states function in the valence band

$$p_0 = N_v \exp \left[\frac{-(E_F - E_v)}{kT} \right]$$

Thermal-Equilibrium Carrier Concentration

$$N_c = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2}$$

$$N_v = 2 \left(\frac{2\pi m_p^* kT}{h^2} \right)^{3/2}$$

$$n_0 = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$

$$p_0 = N_v \exp \left[\frac{-(E_F - E_v)}{kT} \right]$$

	$N_c(\text{cm}^{-3})$	$N_v(\text{cm}^{-3})$	m_n^*/m_0	m_p^*/m_0
Si	2.8×10^{19}	1.04×10^{19}	1.08	0.56
Gallium Arsenide	4.7×10^{17}	7.0×10^{18}	0.067	0.48
Germanium	1.04×10^{19}	6.0×10^{18}	0.55	0.37

If E_F is closer to E_c , $n > p$
 If E_F is closer to E_v , $n < p$
 If E_F is in the midgap, $n = p$

The effective density of states N_c in the conduction band or the valence band N_v is the density of electrons in the conduction band or holes in the valence band when the Fermi level coincides with the conduction band edge E_c or the valence band edge E_v .

Intrinsic semiconductor

Intrinsic electron concentration = Intrinsic hole concentration

$$n_i = p_i$$

Intrinsic carrier concentration

Why?

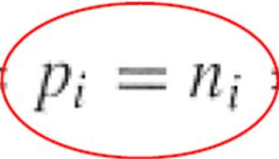
- charge carriers due to thermal excitation
- thermally generated electrons and holes always created in pairs.

The Intrinsic Carrier Concentration

$$n_0 = n_i = N_c \exp \left[\frac{-(E_c - E_{Fi})}{kT} \right]$$

Intrinsic Fermi level



$$p_0 = p_i = n_i = N_v \exp \left[\frac{-(E_{Fi} - E_v)}{kT} \right]$$


Intrinsic carrier concentration

$$n_i^2 = N_c N_v \exp \left[\frac{-(E_c - E_{Fi})}{kT} \right] \cdot \exp \left[\frac{-(E_{Fi} - E_v)}{kT} \right]$$

The Intrinsic Carrier Concentration

$$n_i^2 = N_c N_v \exp \left[\frac{-(E_c - E_v)}{kT} \right] = N_c N_v \exp \left[\frac{-E_g}{kT} \right]$$

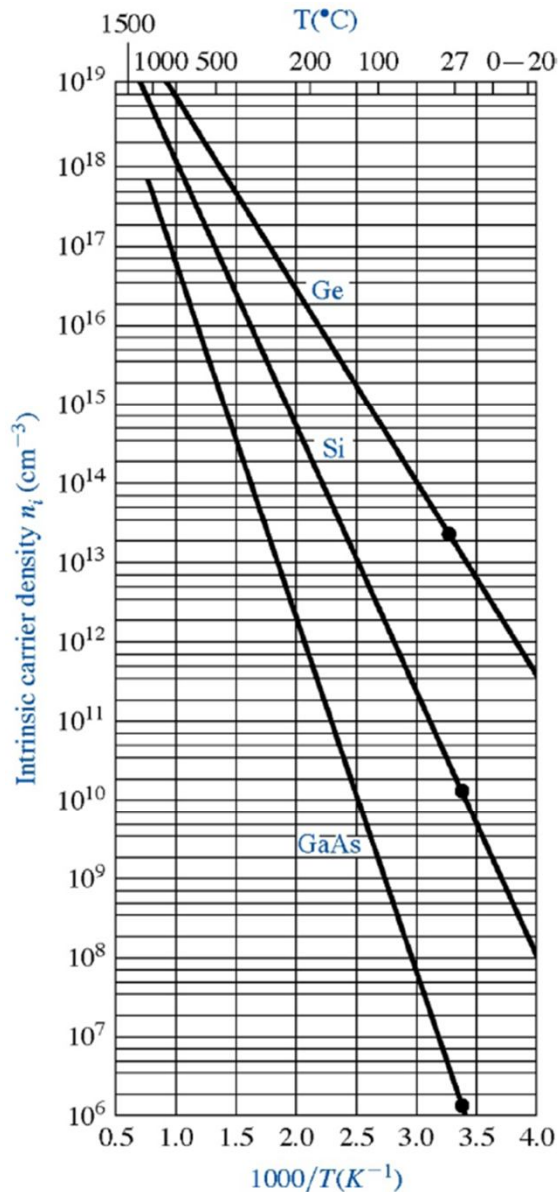
Dependent on temperature and band gap.

Commonly accepted values of n_i at $T=300\text{K}$

Semiconductor	N_i
Silicon	$1.5 \times 10^{10} \text{ cm}^{-3}$
Gallium Arsenide	$1.8 \times 10^6 \text{ cm}^{-3}$
Germanium	$2.4 \times 10^{13} \text{ cm}^{-3}$

We can think of the intrinsic concentration n_i simply as a parameter of the semiconductor material.

The Intrinsic Carrier Concentration



$$n_i^2 = N_c N_v \exp \left[\frac{-(E_c - E_v)}{kT} \right] = N_c N_v \exp \left[\frac{-E_g}{kT} \right]$$

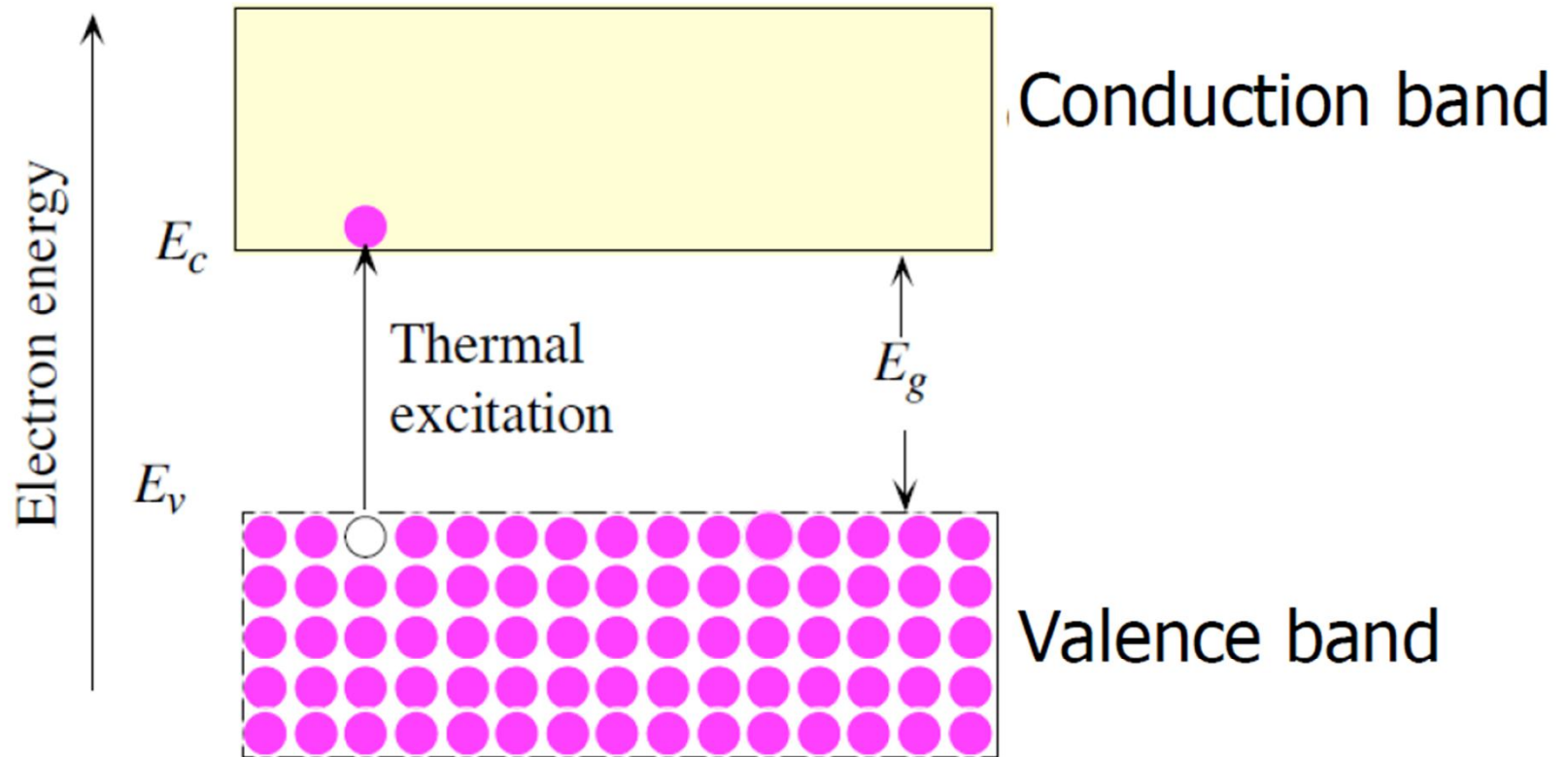
$$\log_{10}(n_i) = \frac{1}{2} \log_{10}(N_c N_v) - \frac{E_g}{2kT} \log_{10} e$$

Plot of $\log_{10}(n_i)$ vs. $1/T$ is straight line.

Slope is negative. From the slope E_g can be calculated

Dependent on temperature and band gap.

Where is the intrinsic Fermi level?



E_{Fi} (Intrinsic Fermi level): E_F at which electron and hole concentration becomes equal

Where is the intrinsic Fermi level?

Electron concentration

Hole concentration

$$N_c \exp \left[\frac{-(E_c - E_{Fi})}{kT} \right] = N_v \exp \left[\frac{-(E_{Fi} - E_v)}{kT} \right]$$

$$E_{Fi} = \frac{1}{2}(E_c + E_v) + \frac{1}{2} kT \ln \left(\frac{N_v}{N_c} \right)$$

$$E_{Fi} = \frac{1}{2}(E_c + E_v) + \frac{3}{4} kT \ln \left(\frac{m_p^*}{m_n^*} \right)$$

$$N_c = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2}$$

$$\frac{1}{2}(E_c + E_v) = E_{\text{midgap}} \quad \longrightarrow$$

$$E_{Fi} - E_{\text{midgap}} = \frac{3}{4} kT \ln \left(\frac{m_p^*}{m_n^*} \right)$$

Even in intrinsic semiconductor, Fermi level is not exactly at centre between conduction and valence bands.

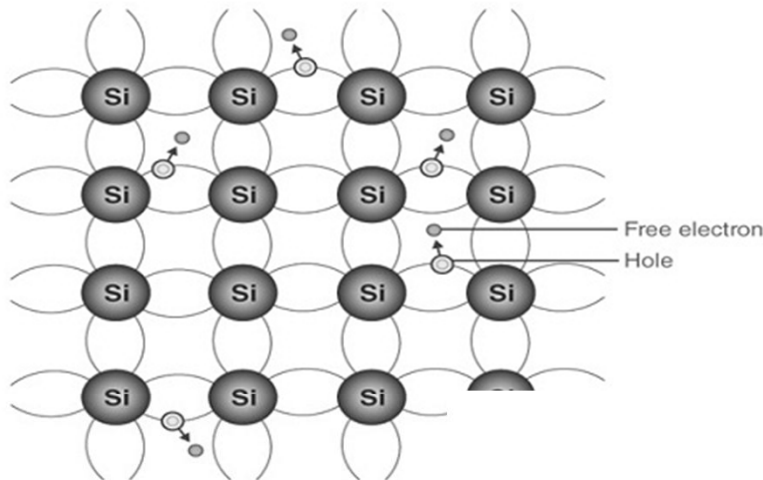
Where is the intrinsic Fermi level?

$$E_{Fi} - E_{\text{midgap}} = \frac{3}{4} kT \ln \left(\frac{m_p^*}{m_n^*} \right)$$

If the electron and hole effective masses are equal so that $m_p^* = m_n^*$, then the intrinsic Fermi level is exactly in the center of the bandgap. If $m_p^* > m_n^*$, the intrinsic Fermi level is slightly above the center, and if $m_p^* < m_n^*$, it is slightly below the center of the bandgap. The density of states function is directly related to the carrier effective mass; thus, a larger effective mass means a larger density of states function. The intrinsic Fermi level must shift away from the band with the larger density of states in order to maintain equal numbers of electrons and holes.

Intrinsic Vs. Extrinsic semiconductor

intrinsic



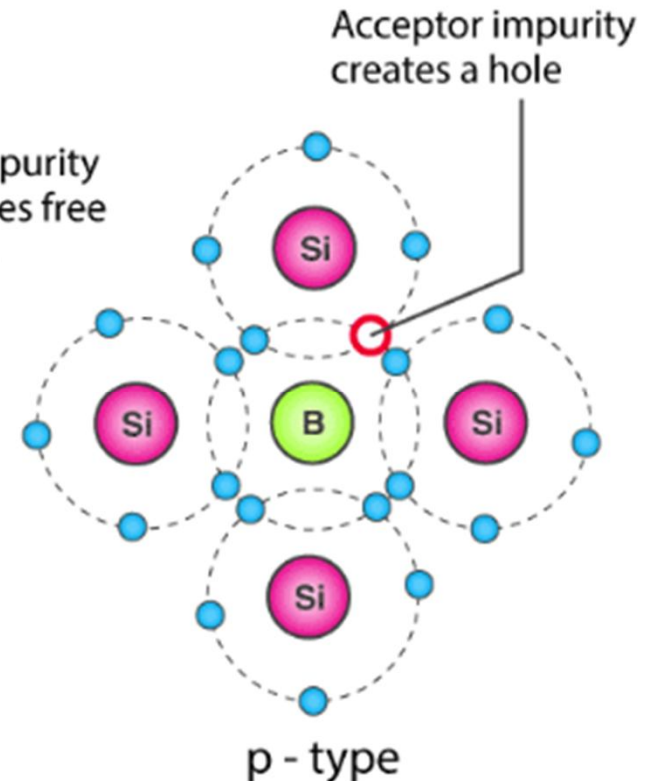
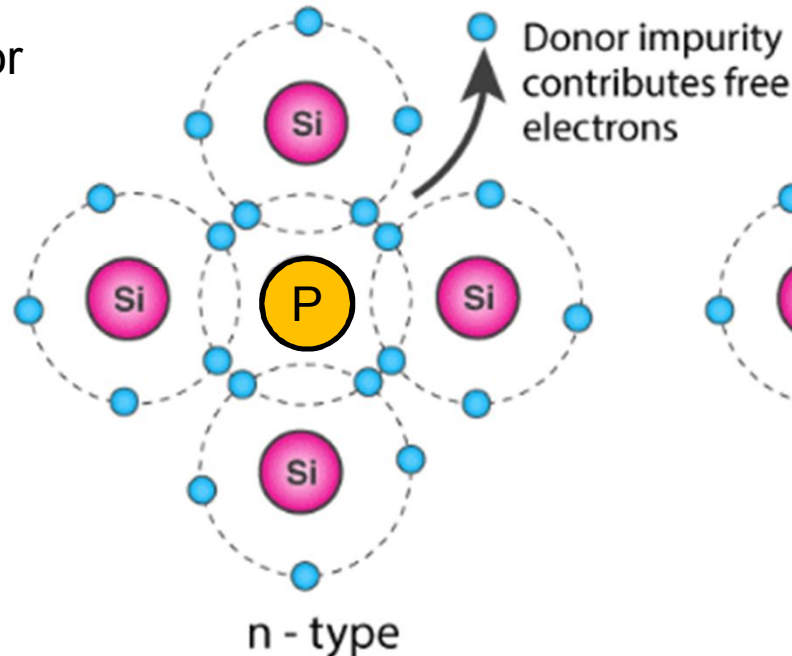
Extrinsic semiconductor

- Also called as impurity semiconductor
- atom with more than 4 valence electrons: donor impurity
- less than 4: acceptor impurity.

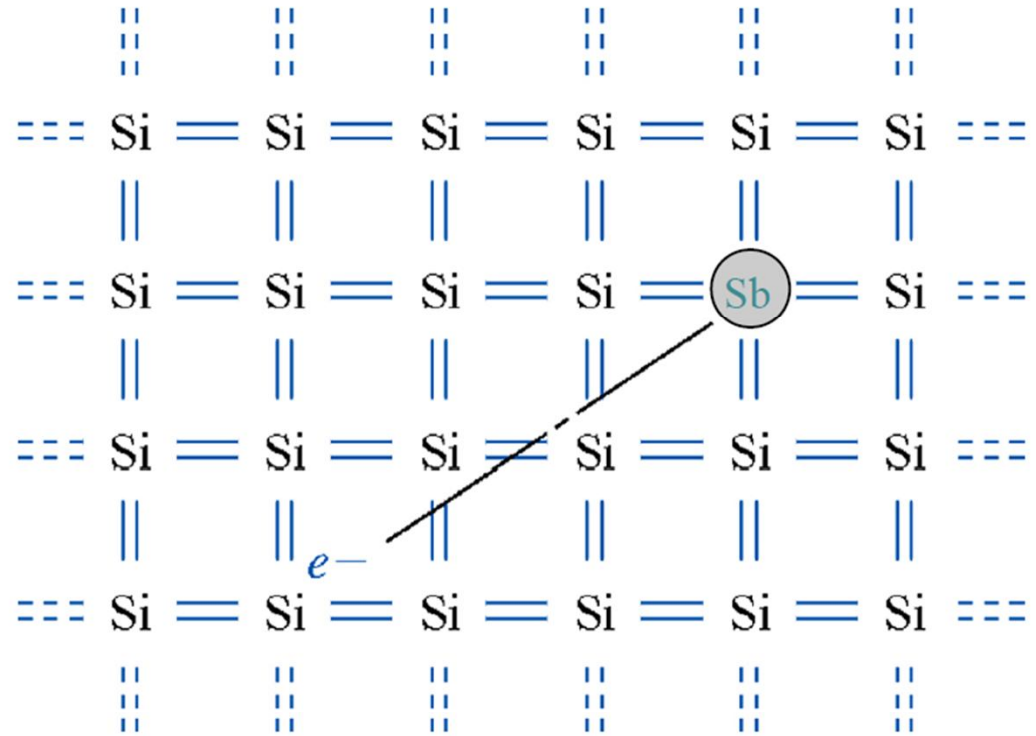
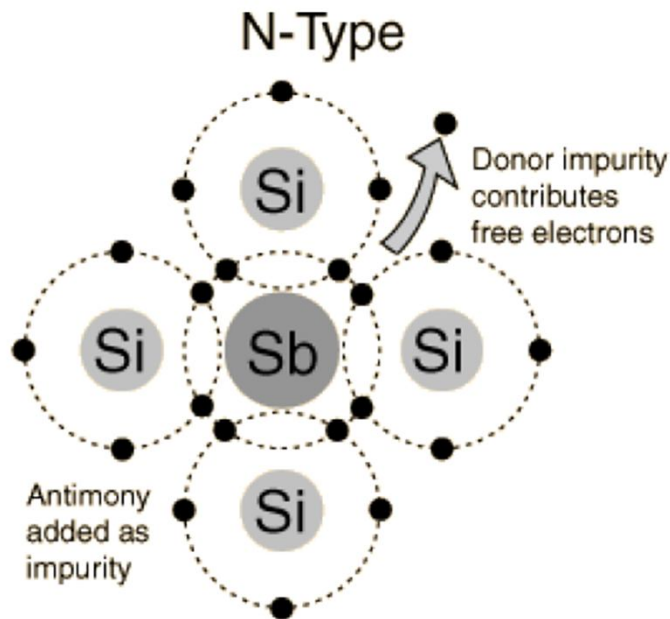
acceptor

IIIA	IVA	VA
13	14	15
5 B 硼 10.81	6 C 碳 12.01	7 N 氮 14.01
13 Al 铝 26.98	14 Si 硅 28.09	15 P 磷 30.97
31 Ga 镓 69.72	32 Ge 锗 72.64	33 As 砷 74.92
49 In 铟 114.8	50 Sn 锡 118.7	51 Sb 锑 121.8
81 Tl 铊 204.4	82 Pb 铅 207.2	83 Bi 铋 209.0

donor



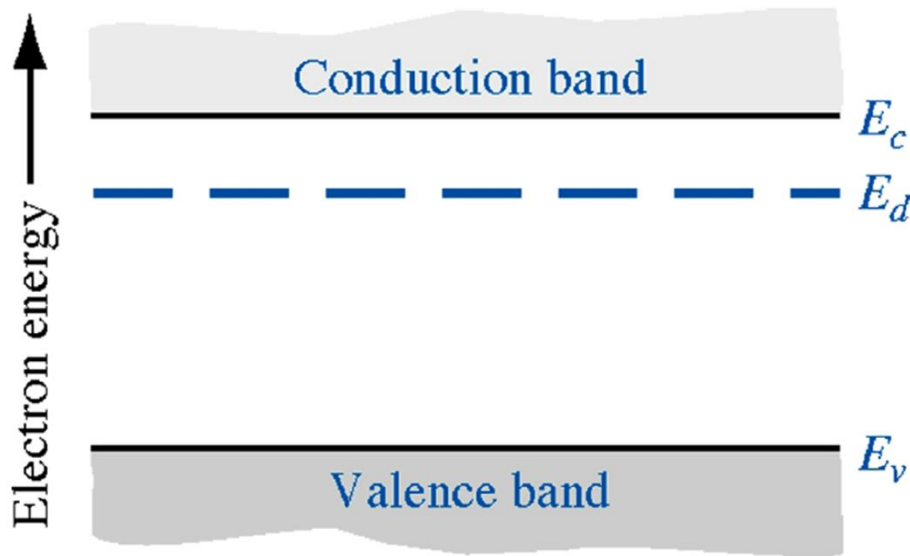
Donor Impurity



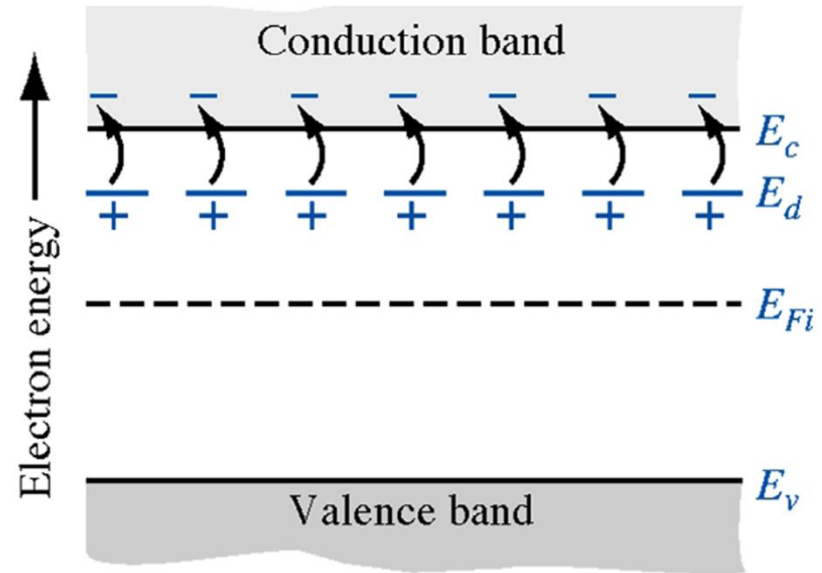
Silicon lattice doped with donor impurity

- At very low temperature, the donor (excess) electron is still bound to the impurity atom.
- However, the donor electron is loosely bound to the impurity atom and can become free with small amount of thermal energy. Impurity atom is then ionized and positively charged.

Donor electron energy level



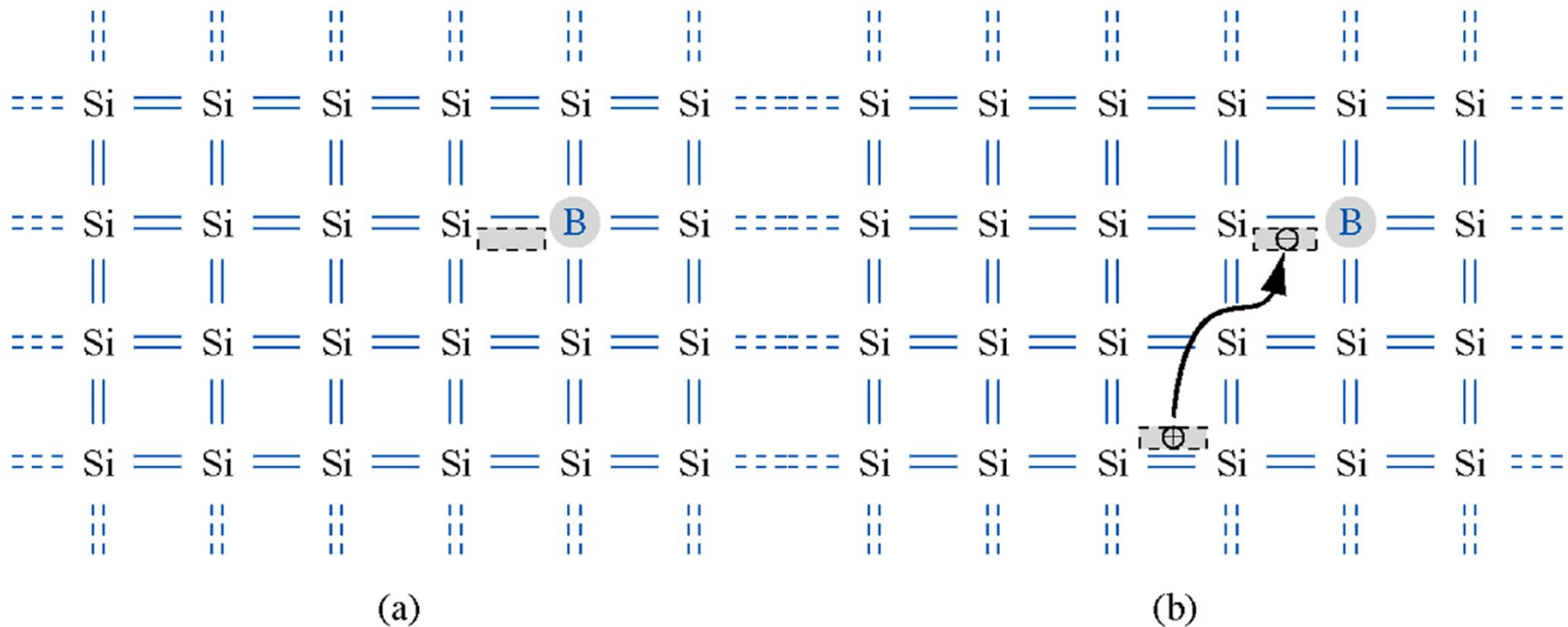
(a)



(a)

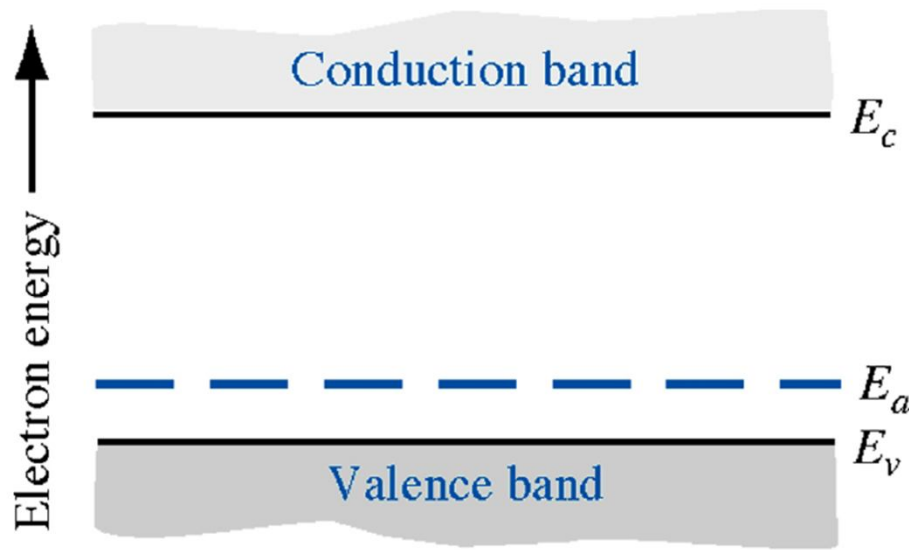
- little energy required to move donor electrons from donor states to conduction band.
- positively charged donor ions are fixed but donor electrons in the conduction band can move through the crystal.

Acceptor Impurity

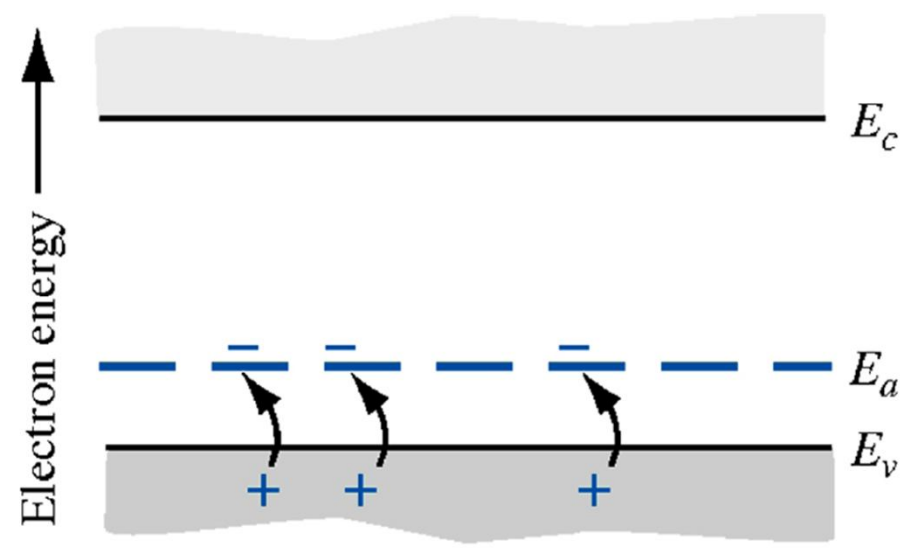


- One covalent bond is incomplete for Si.
- With little thermal energy, a valence electron can break from another covalent bond and can occupy this position, thus creating a hole at the location of the broken covalent bond.
- The acceptor impurity is then ionized and negatively charged.

Acceptor Energy Level



(a)



(b)

- little energy required to move valence electrons to acceptor levels.
- negatively charged acceptor ions are fixed but holes in the valence band can move through the crystal.

Electron concentration vs. temperature in n-type semiconductor

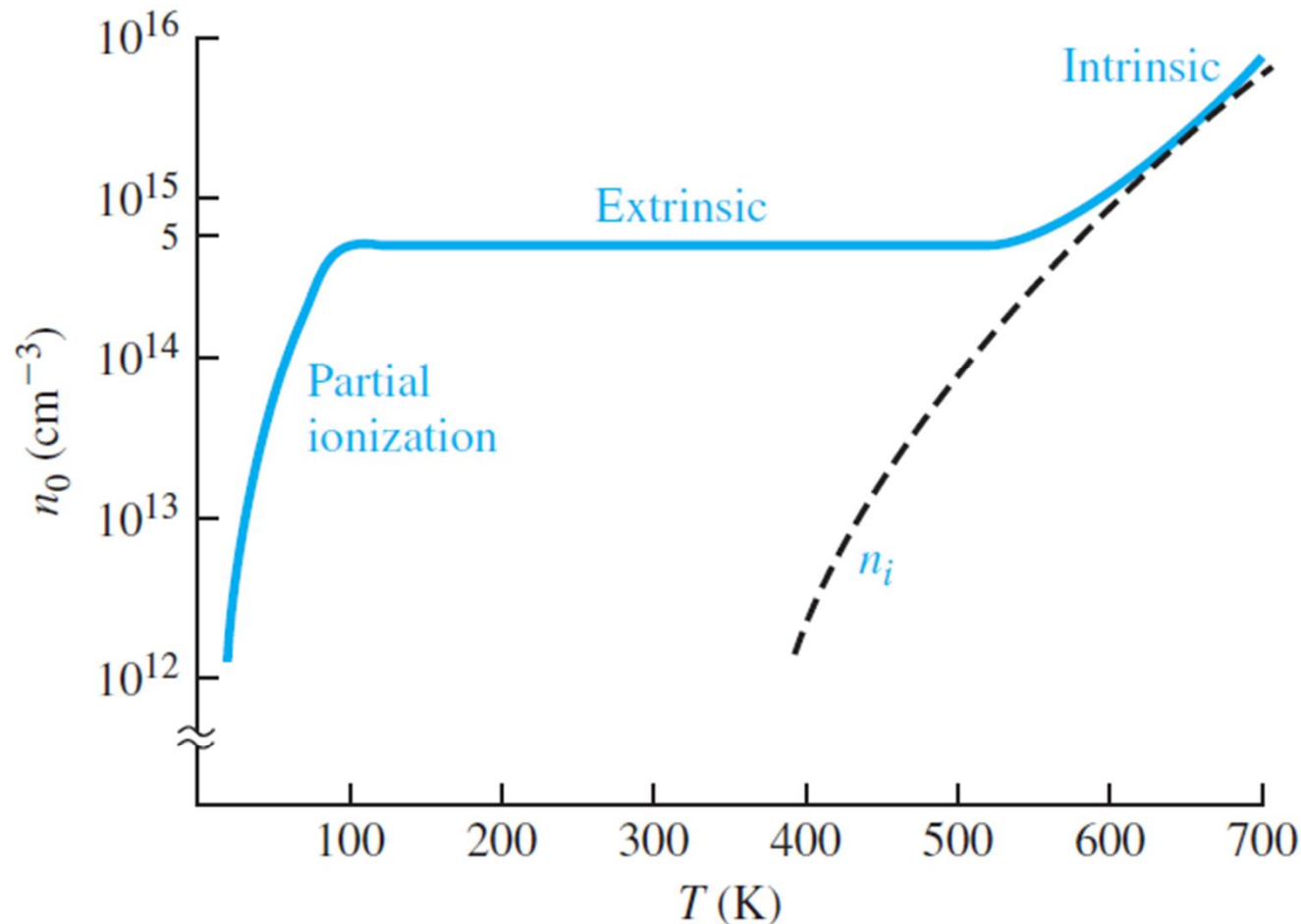
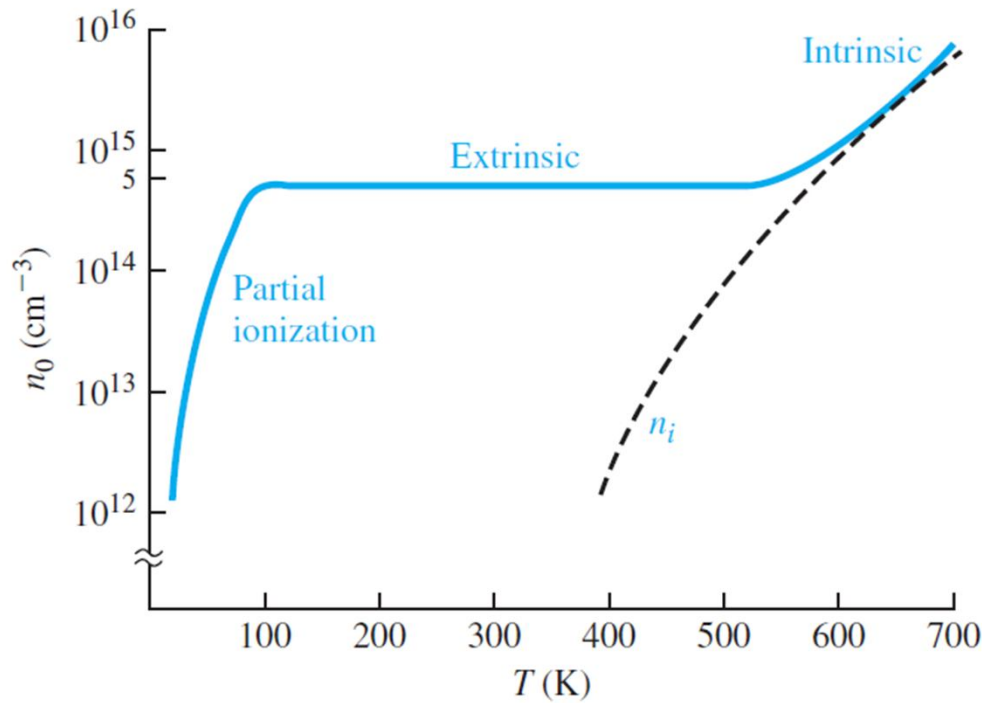


Figure 4.16 | Electron concentration versus temperature showing the three regions: partial ionization, extrinsic, and intrinsic.

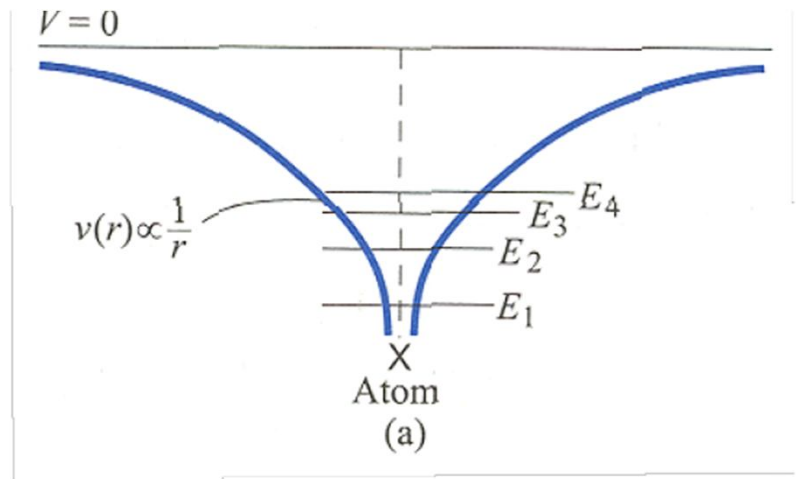
Electron concentration vs. temperature in n-type semiconductor



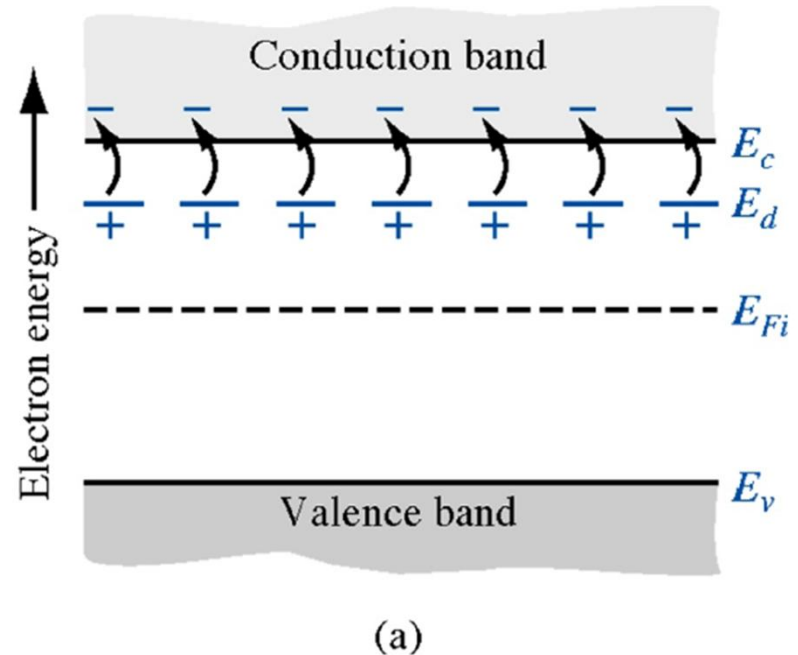
- At low temperatures, donor impurities are partially ionized. As temperature increases the percentage of ionized donor impurities also increases
- Once all donor impurities are ionized, there is no increase in carrier concentration. Even though intrinsic carrier concentration continues to increase, it is still small compared to extrinsic concentration.
- At high temperatures, intrinsic carrier concentration dominates and electron concentration continues to increase again.

Ionization energy

The ionization energy is the energy necessary to remove an electron from the neutral atom.



In case of donor atoms, the ionization energy is the energy necessary to elevate an electron from the donor level to conduction band.



Ionization energy

- In the next few slides, we will calculate the approximate ionization energy for donor atoms.
- We use Bohr atomic model for these calculations. For hydrogen atom, Bohr model and quantum mechanics give similar results.
- Donor impurity atom can be visualized as one donor electron orbiting the positively charged donor ion. This condition is similar to that in a hydrogen atom.
- However we have to consider permittivity of silicon instead of permittivity of free space.

Angular Momentum Quantization

Bohr proposed that circumference of electron orbit = integer number of wavelengths $\rightarrow 2\pi r = n\lambda_n$

Then angular momentum,
$$L = mvr = \frac{hr}{\lambda} = \frac{hr}{\left[\frac{2\pi r}{n}\right]} = \frac{nh}{2\pi}$$

angular momentum of electron is quantized.

Ionization energy calculation

Coulomb attraction force

Centripetal force

$$\frac{e^2}{4\pi\epsilon r_n^2} = \frac{m^* v^2}{r_n}$$

Angular momentum quantization $\longrightarrow m^* r_n v = n\hbar$

$$r_n = \frac{n^2 \hbar^2 4\pi\epsilon}{m^* e^2}$$

Ionization energy calculation

As defined in Chapter 2,
Bohr radius = $a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_0e^2} = 0.53 \text{ \AA}$

$$\frac{\text{orbiting electron radius}}{\text{Bohr radius}} = \frac{r_n}{a_0} = n^2\epsilon_r \left(\frac{m_0}{m^*} \right)$$

For silicon,

$$\epsilon_r = 11.7 \text{ and } \frac{m^*}{m_0} = 0.26$$

$$\text{For } n=1, \quad \frac{r_1}{a_0} = 45$$

Ionization energy calculation

- $r_1/a_0=45$ or $r_1=23.9\text{\AA}$
- This radius ~ 4 lattice constants of Si.
- Each unit cell contains 8 silicon atoms.
- Donor electron thus loosely bound to the donor atom.
- We will next find the approximate ionization energy.

Ionization energy calculation

Total energy

$$E = T + V$$


Kinetic energy

Potential energy

$$T = \frac{1}{2}m^*v^2 \quad \longrightarrow \quad T = \frac{m^*e^4}{2(n\hbar)^2(4\pi\epsilon)^2}$$

$$V = \frac{-e^2}{4\pi\epsilon r_n} = \frac{-m^*e^4}{(n\hbar)^2(4\pi\epsilon)^2}$$

$$E = T + V = \frac{-m^*e^4}{2(n\hbar)^2(4\pi\epsilon)^2}$$

Ionization energy

- Ionisation energy of Hydrogen in lowest energy state = -13.6eV
- For Si, it is $-25.8\text{meV} \ll \text{band gap}$.
- Calculations using Bohr model give only the order of magnitude of the ionisation energy. Actual values differ.

Table 4.3 | Impurity ionization energies in silicon and germanium

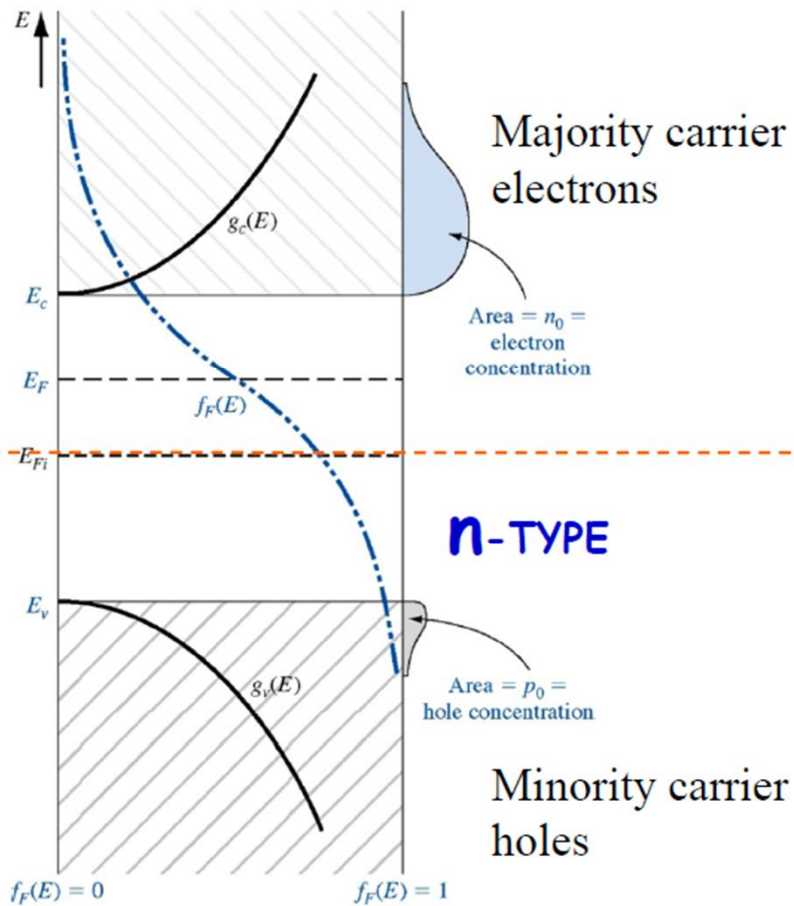
Impurity	Ionization energy (eV)	
	Si	Ge
<i>Donors</i>		
Phosphorus	0.045	0.012
Arsenic	0.05	0.0127
<i>Acceptors</i>		
Boron	0.045	0.0104
Aluminum	0.06	0.0102

Table 4.4 | Impurity ionization energies in gallium arsenide

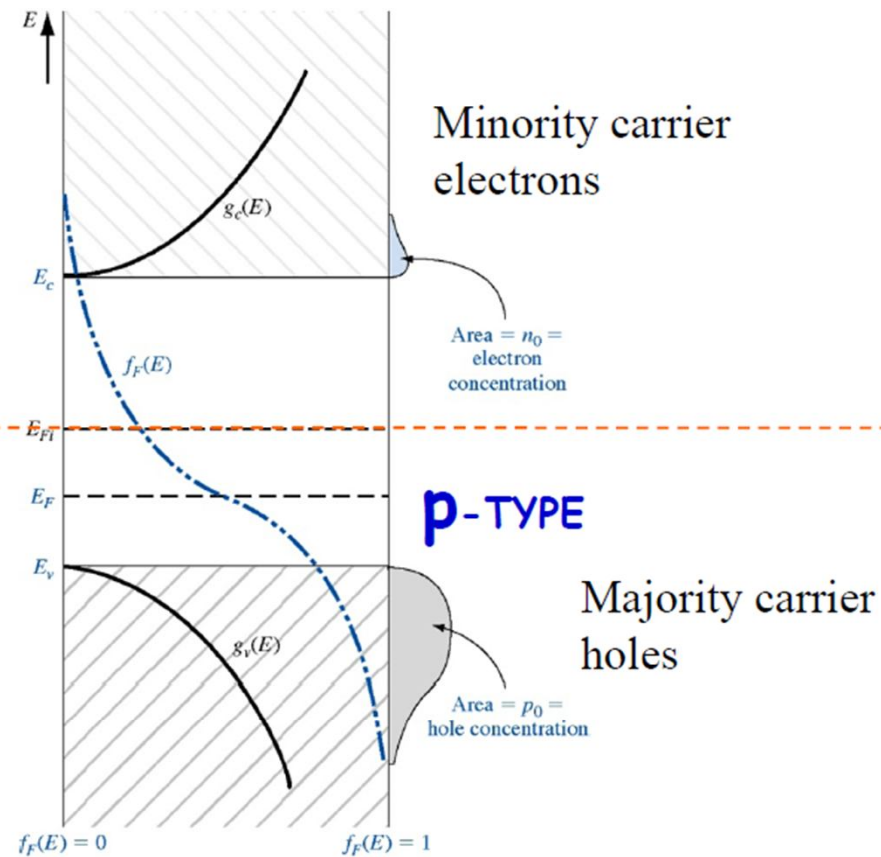
Impurity	Ionization energy (eV)
<i>Donors</i>	
Selenium	0.0059
Tellurium	0.0058
Silicon	0.0058
Germanium	0.0061
<i>Acceptors</i>	
Beryllium	0.028
Zinc	0.0307
Cadmium	0.0347
Silicon	0.0345
Germanium	0.0404

$$kT = 0.026 \text{ eV}$$

4.3 | THE EXTRINSIC SEMICONDUCTOR



$$n_0 = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$



$$p_0 = N_v \exp \left[\frac{-(E_F - E_v)}{kT} \right]$$

Carrier concentration

$$n_0 = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$

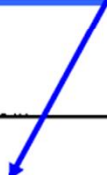
$$p_0 = N_v \exp \left[\frac{-(E_F - E_v)}{kT} \right]$$

The equation is valid for both intrinsic and extrinsic semiconductors

Another form (relation between E_F and E_{Fi})

Intrinsic carrier concentration

$$n_0 = N_c \exp\left[\frac{-(E_c - E_{Fi})}{kT}\right] \exp\left[\frac{(E_F - E_{Fi})}{kT}\right]$$


$$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]$$

$$p_0 = n_i \exp\left[\frac{-(E_F - E_{Fi})}{kT}\right]$$

$$n_0 p_0 = n_i^2$$

- one of the fundamental principles of semiconductors in thermal equilibrium.
- This equation was derived using the Boltzmann approximation. If the Boltzmann approximation is not valid, then likewise, this equation is not valid.

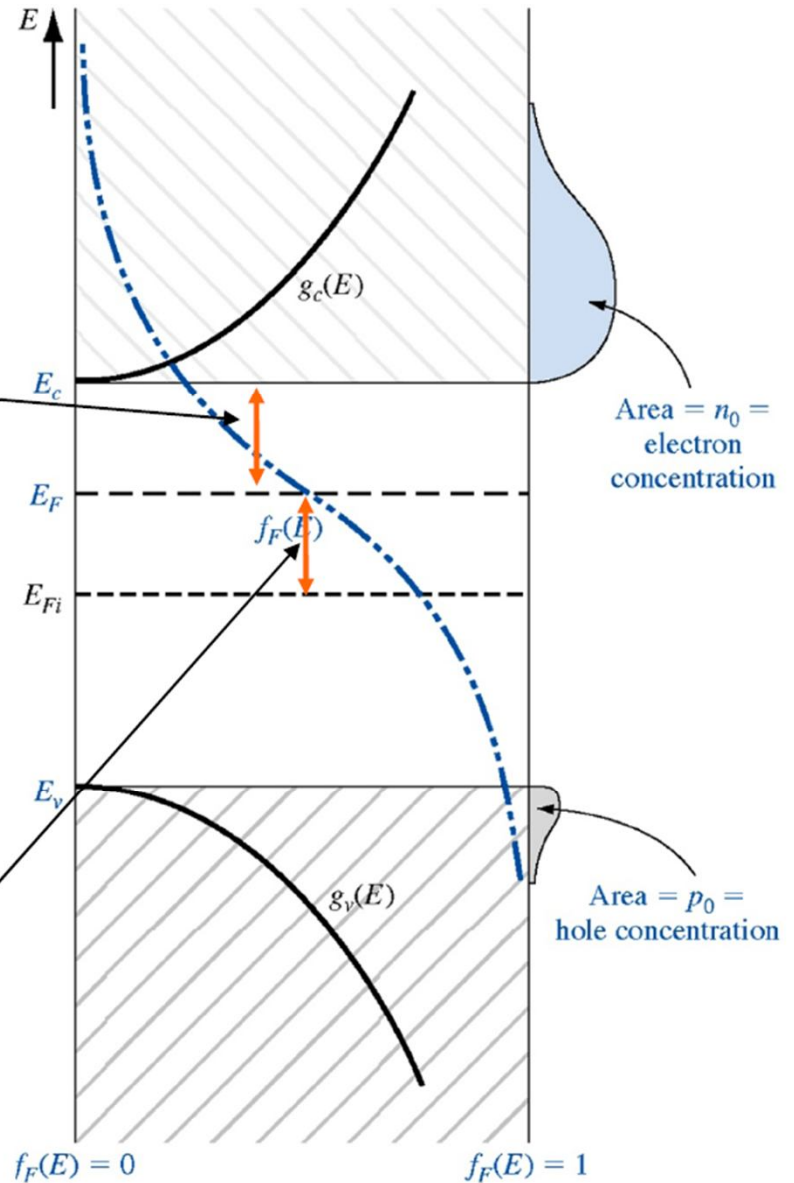
Where is the Fermi level?

$$n_0 = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$

$$E_c - E_F = kT \ln \left(\frac{N_c}{n_0} \right)$$

$$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right]$$

$$E_F - E_{Fi} = kT \ln \left(\frac{n_0}{n_i} \right)$$



Where is the Fermi level?

Example: consider silicon at $T = 300$ K so that $N_c = 2.8 \times 10^{19} \text{ cm}^{-3}$ and $N_v = 1.04 \times 10^{19} \text{ cm}^{-3}$. If we assume the Fermi energy is 0.25 eV below the conduction band, calculate the thermal equilibrium concentrations of electrons and holes. The bandgap energy of silicon is 1.12 eV.

Solution: $E_c - E_F = 0.25 \text{ eV}$ $E_F - E_v = 0.87 \text{ eV}$

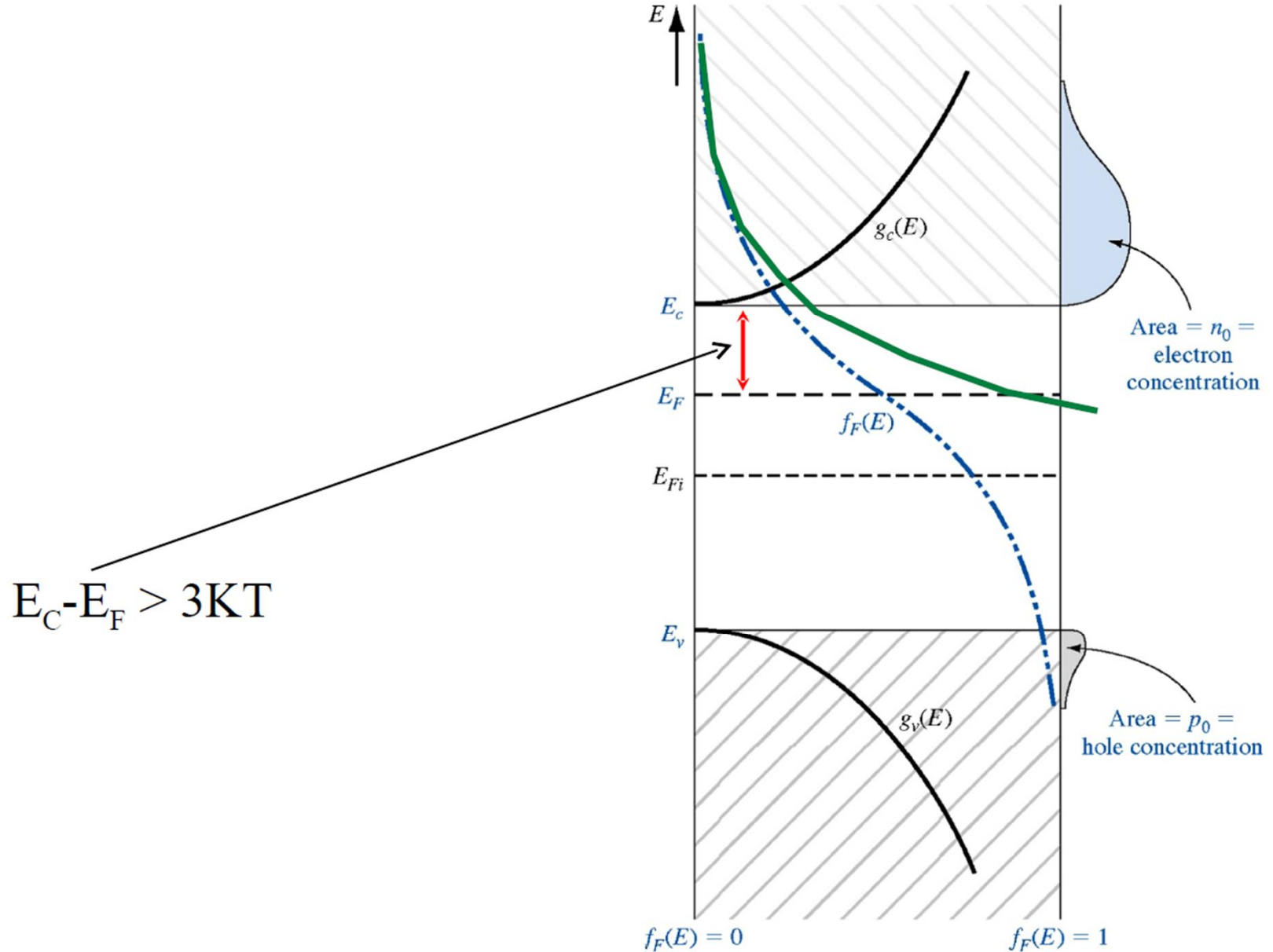
$$n_0 = (2.8 \times 10^{19}) \exp \left[\frac{-(0.25)(1.602 \times 10^{-19})}{(1.381 \times 10^{-23})(300)} \right] = 1.8 \times 10^{15} \text{ cm}^{-3}$$

$$p_0 = (1.04 \times 10^{19}) \exp \left[\frac{-(0.87)(1.602 \times 10^{-19})}{(1.381 \times 10^{-23})(300)} \right] = 2.6 \times 10^4 \text{ cm}^{-3}$$

Comment: electron and hole concentrations change by orders of magnitude from the intrinsic carrier concentrations (at 300 K, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$) as the Fermi energy changes by a few tenths of an eV.

In an n-type semiconductor, $n_0 > p_0$, electrons are referred to as **majority carriers** and holes as **minority carriers**. In a p-type semiconductor, $p_0 > n_0$, holes are referred to as **majority carriers** and electrons as **minority carriers**.

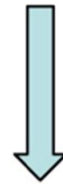
Condition for the Boltzmann approximation



If the impurity concentration is very high....

Fermi level will be very close to conduction band or valence band.

No Boltzmann approximation



Use The Fermi-Dirac Integral

If the impurity concentration is very high....

$$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

Boltzman approximation

$$f_F(E) \approx \exp\left[\frac{-(E - E_F)}{kT}\right]$$

$$n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Only if $E_c - E_F > 3kT$

If the impurity concentration is very high....

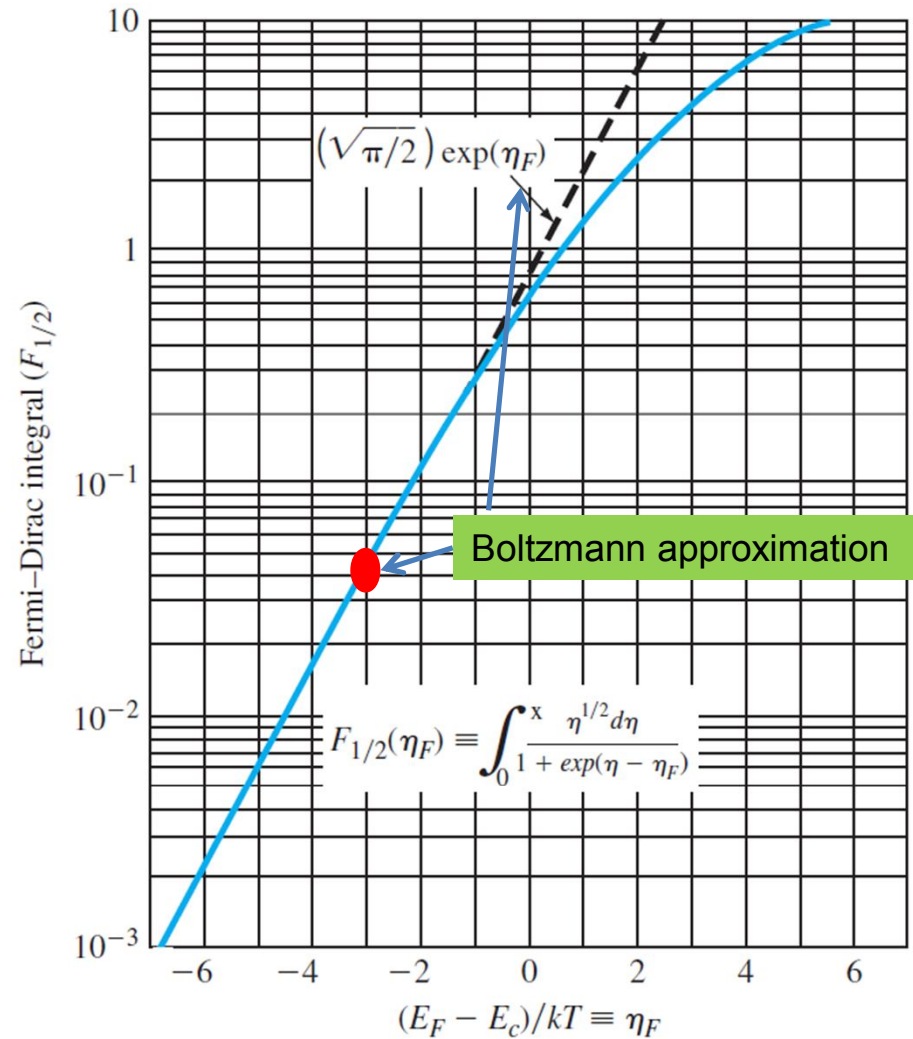
Fermi-Dirac Integral

$$n_0 = \frac{4\pi}{h^3} (2m_n^*)^{3/2} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2} dE}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$$

$$\eta = \frac{E - E_c}{kT} \quad \eta_F = \frac{E_F - E_c}{kT}$$

$$n_0 = 4\pi \left(\frac{2m_n^* kT}{h^2} \right)^{3/2} \int_0^{\infty} \frac{\eta^{1/2} d\eta}{1 + \exp(\eta - \eta_F)}$$

$$F_{1/2}(\eta_F) = \int_0^{\infty} \frac{\eta^{1/2} d\eta}{1 + \exp(\eta - \eta_F)}$$



If $\eta_F > 1$, then $E_F > E_c$

Degenerate Semiconductors

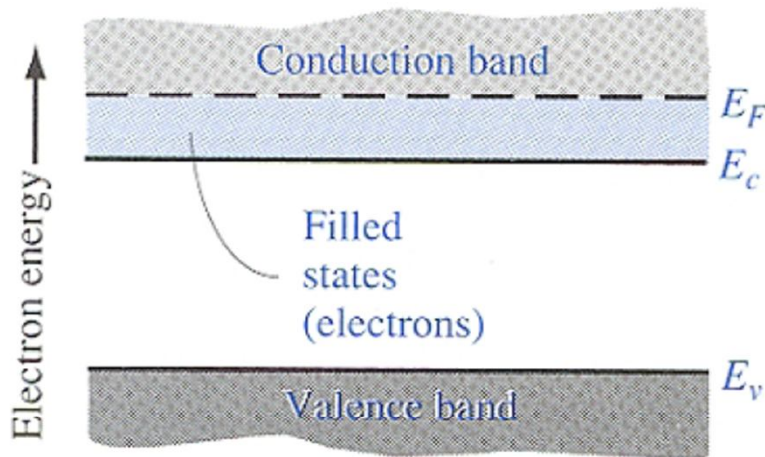
If the impurity atoms are very close each other...

- Donor electrons interact with each other
- The single discrete donor energy will split into a band
- The band may overlap the conduction band
- If the concentration exceed N_c , E_F lies within the conduction band

$np = n_i^2$ and $n = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$ no longer valid, n can reach saturation of $10^{20}/\text{cm}^3$

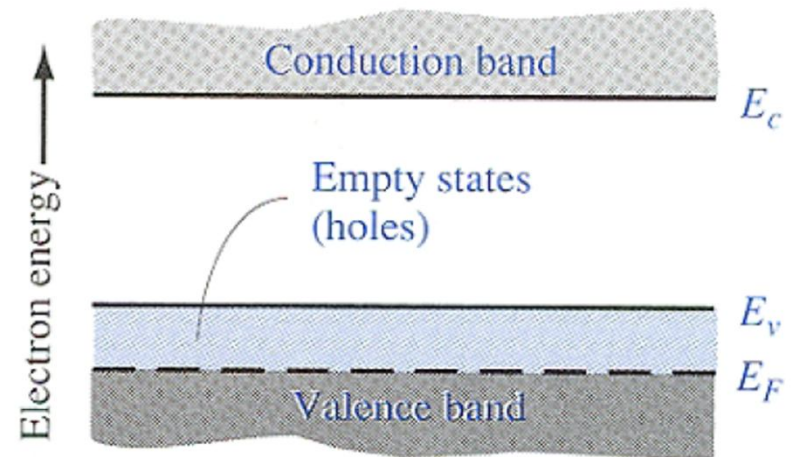
Degenerate Semiconductors

$$N_d > N_c$$



(a)

$$N_a > N_v$$



(b)

Fermi level in the conduction band: Metallic conduction

4.4 | STATISTICS OF DONORS AND ACCEPTORS

How many electrons still in the donor levels compared to the total number of electrons?

depends on the temperature and the Fermi level....

Probability function for donor & acceptor levels

Density of electrons
occupying donor states

Density of donor atoms

$$\frac{n_d}{N_d} = \frac{1}{1 + \frac{1}{2} \exp\left(\frac{E_d - E_F}{kT}\right)}$$

By adding one electron, the vacancy requirement of the atom is satisfied, and the addition of a second electron in the donor level is not possible.

This same as the Fermi-Dirac probability function except the pre-exponential coefficient of $\frac{1}{2}$.

$$n_d = N_d - N_d^+$$

Concentration of ionized donors

Probability function for donor & acceptor levels

similar for holes:

$$p_a = \frac{N_a}{1 + \frac{1}{g} \exp\left(\frac{E_F - E_a}{kT}\right)} = N_a - N_a^-$$

g =degeneration factor; 4 for GaAs and Si acceptor levels

The ground state degeneracy factor g is normally taken as 4 for the acceptor level in silicon and gallium arsenide because of the detailed band structure.

Probability function for donor & acceptor levels

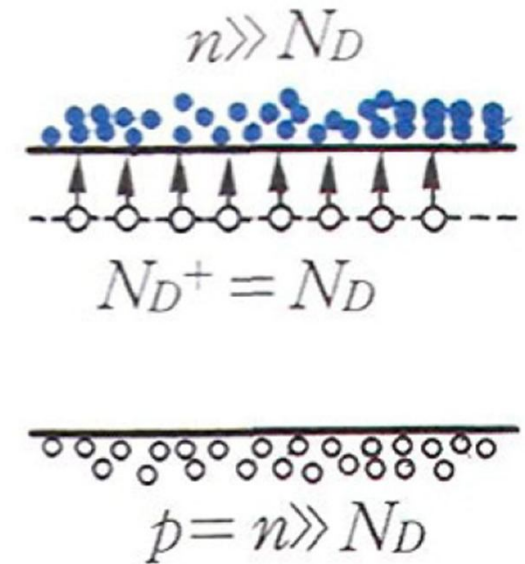
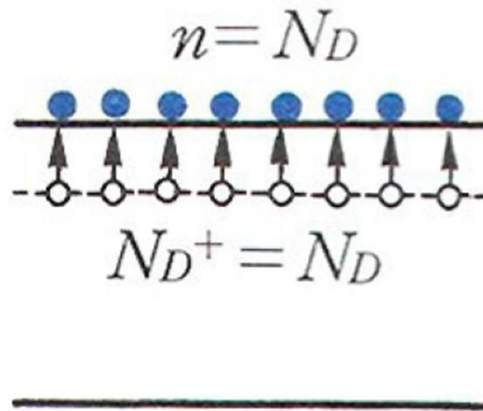
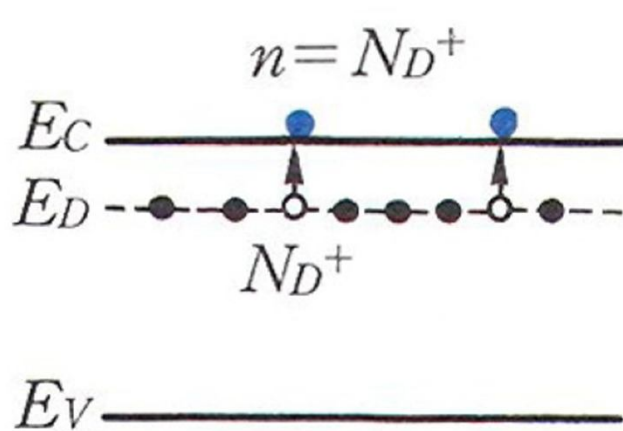
● 能带中的电子是作共有化运动的电子，它们的运动范围延伸到整个晶体，与电子空间运动对应的每个能级，存在自旋相反的两个量子态。由于电子之间的作用很微弱，电子占据这两个量子态是相互独立的。

能带中的电子在状态中的分布是服从费米分布的。

● 杂质上的电子态与上述情形不同，它们是束缚在状态中的局域化量子态。

以类氢施主为例，当基态未被占据时，由于电子自旋方向的不同而可以有两种方式占据状态，但是一旦有一个电子以某种自旋方式占据了该能级，就不再可能有第二个电子占据另一种自旋状态。因为在施主俘获一个电子之后，静电力将把另一个自旋状态提到很高的能量，（因为电子态是局域化的，电子间相互作用很强），基于上述由自旋引起的简并，不能用费米分布函数来确定电子占据施主能级的几率。

How many electrons still in the donor levels



Low temperature



Moderate temperature

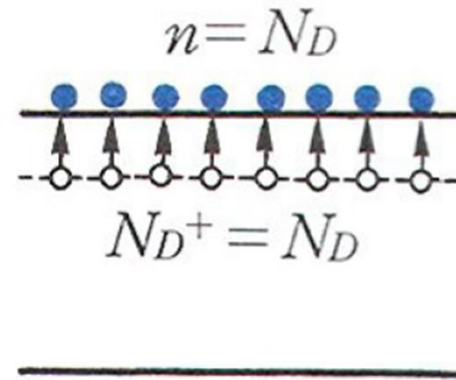


High temperature

Moderate temperature

$$E_d - E_F \gg kT$$

$$\frac{n_d}{N_d} = \frac{1}{1 + \frac{1}{2} \exp\left(\frac{E_d - E_F}{kT}\right)}$$



$$n_d \approx \frac{N_d}{\frac{1}{2} \exp\left(\frac{E_d - E_F}{kT}\right)} = 2N_d \exp\left[\frac{-(E_d - E_F)}{kT}\right]$$

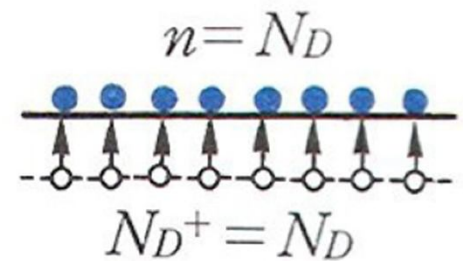
If $E_d - E_F \gg kT$, then even $E_c - E_F \gg kT$

$$\text{Then, } n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Fraction of electrons still in the donor states

$$\frac{n_d}{n_d + n_0} = \frac{2N_d \exp\left[\frac{-(E_d - E_F)}{kT}\right]}{2N_d \exp\left[\frac{-(E_d - E_F)}{kT}\right] + N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]}$$

$$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d} \exp\left[\frac{-(E_c - E_d)}{kT}\right]}$$



With Phosphorus doping of $N_d = 10^{16} \text{cm}^{-3}$, at $T = 300 \text{K}$, $n_d / (n_d + n_0) = 0.41\%$

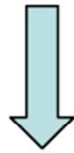
Almost complete ionization at Room Temp!

Extremely low temperature (T=0K)

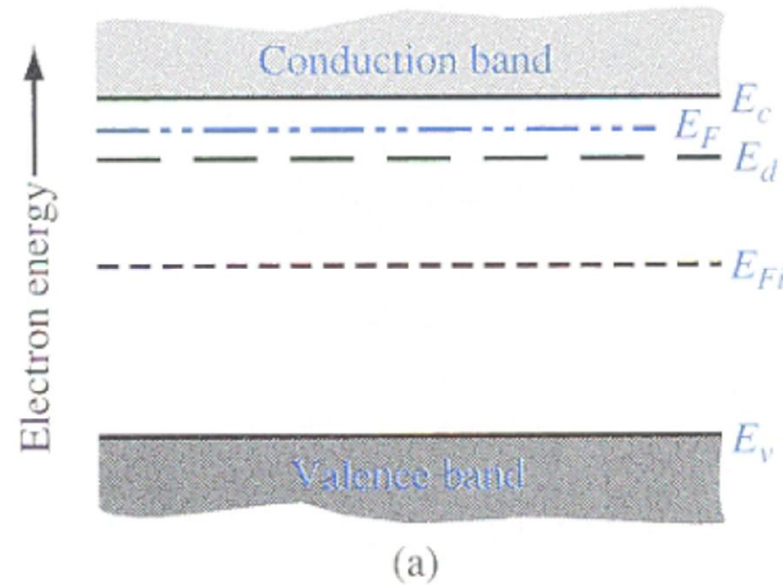
$$n_d = N_d$$

Freeze-out

$$n_d = \frac{N_d}{1 + \frac{1}{2} \exp\left(\frac{E_d - E_F}{kT}\right)}$$

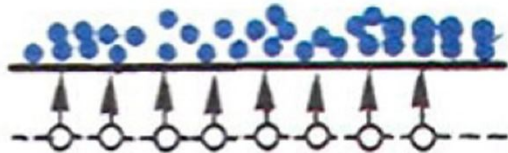


$$E_F > E_d$$



High temperature

$n \gg N_D$ (because of thermally generated electrons)



$$N_D^+ = N_D$$



$p = n \gg N_D$ (because of thermally generated holes)

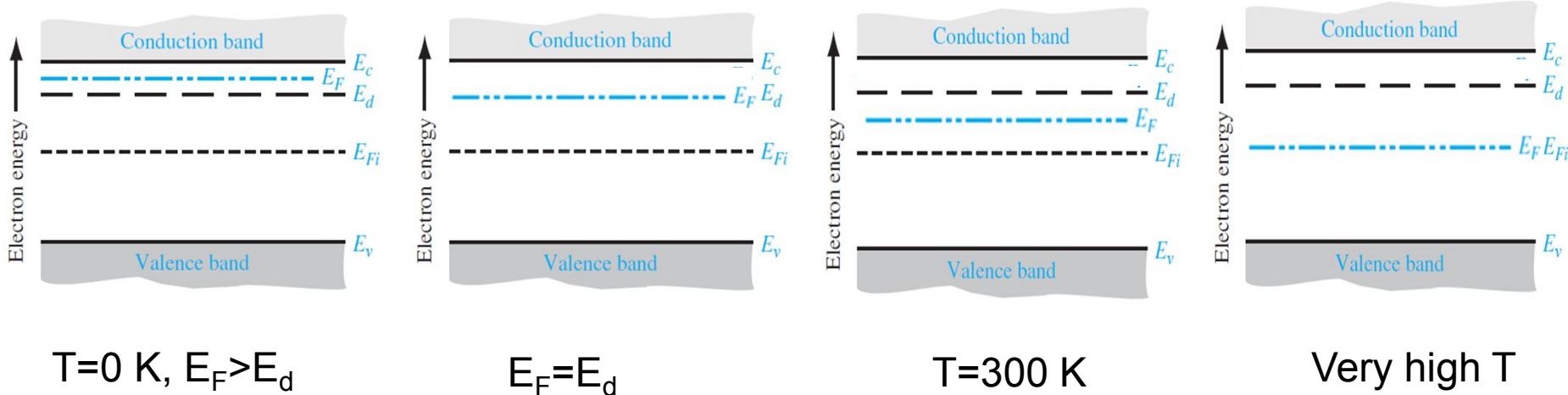
$$n_0 = n_i = N_c \exp \left[\frac{-(E_c - E_{Fi})}{kT} \right]$$

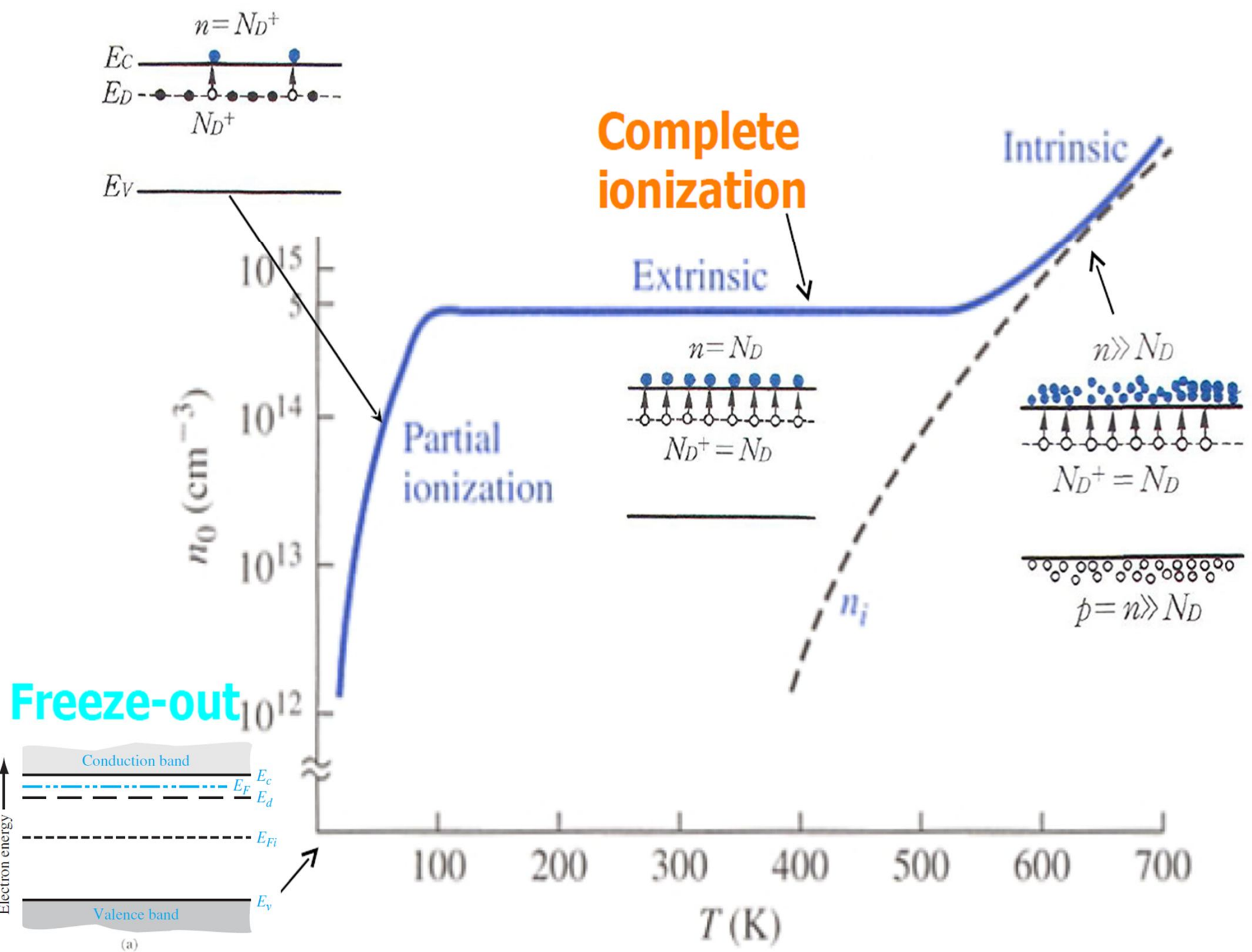
- At very high temperature behavior is just like the intrinsic semiconductor

From low T to high T

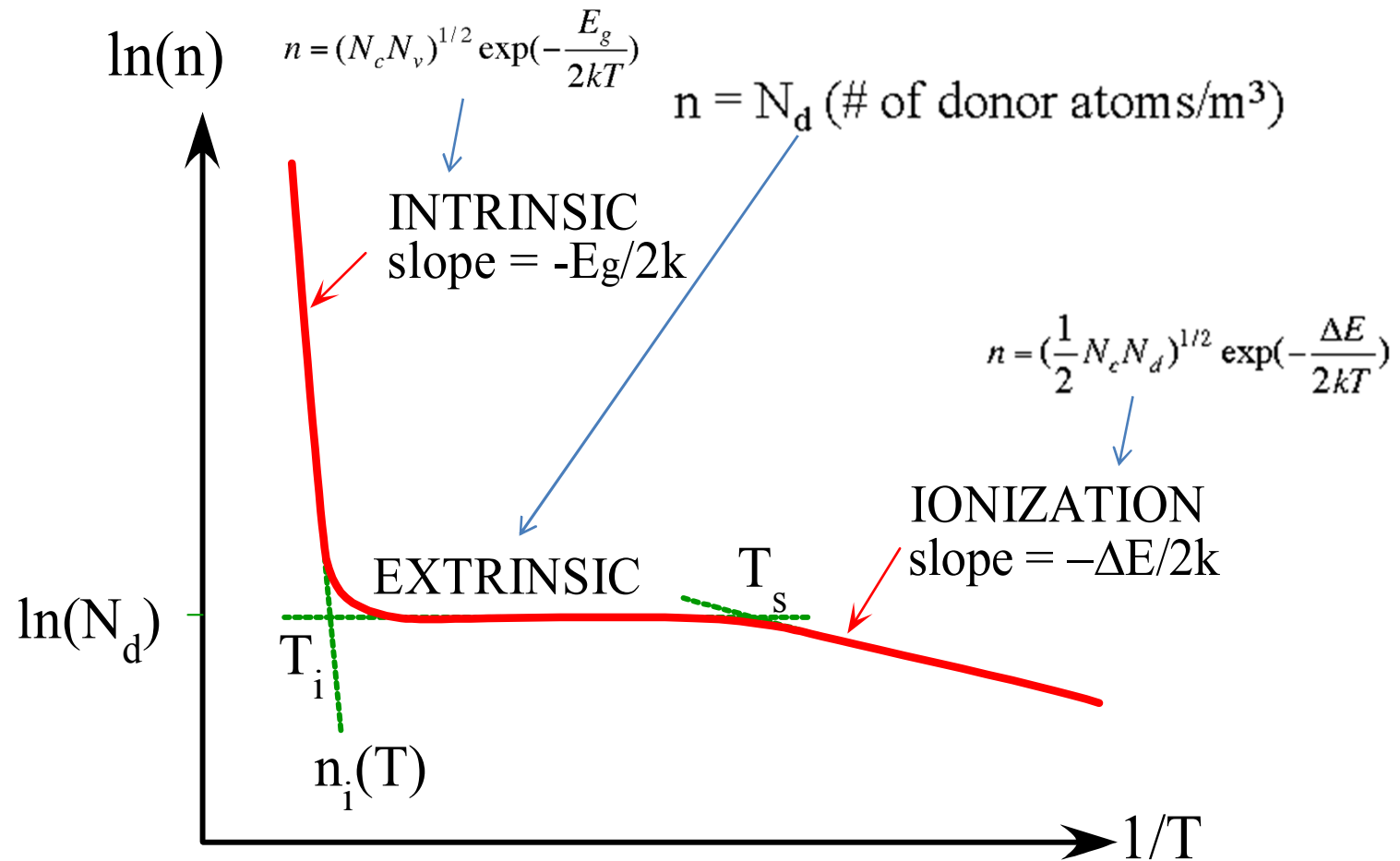
$$n_d = \frac{N_d}{1 + \frac{1}{2} \exp\left(\frac{E_d - E_F}{kT}\right)}$$

- At $T=0\text{K}$, $n_d=N_d$, then $E_F > E_d$
- $T>0\text{K}$, when $E_F=E_d$, $n_d=(2/3) N_d$, thus $N_{d+}=(1/3)N_d$
- $T=300\text{ K}$, $n_d \sim 0$, then $E_F < E_d$, $E_F - E_{Fi} = kT \ln(N_d/n_i)$
- T very high, $n=n_i$, $E_F = E_{Fi}$





Temperature dependence of the electron concentration



The temperature dependence of the electron concentration in an n-type semiconductor.

Compensated semiconductor

- Both donor and acceptor impurities in the same region
- If $N_d > N_a \rightarrow$ n-type compensated semiconductor
- If $N_d < N_a \rightarrow$ p-type compensated semiconductor
- If $N_d = N_a \rightarrow$ completely compensated (will behave like intrinsic material)
- Practical semiconductor is always compensated semiconductor.

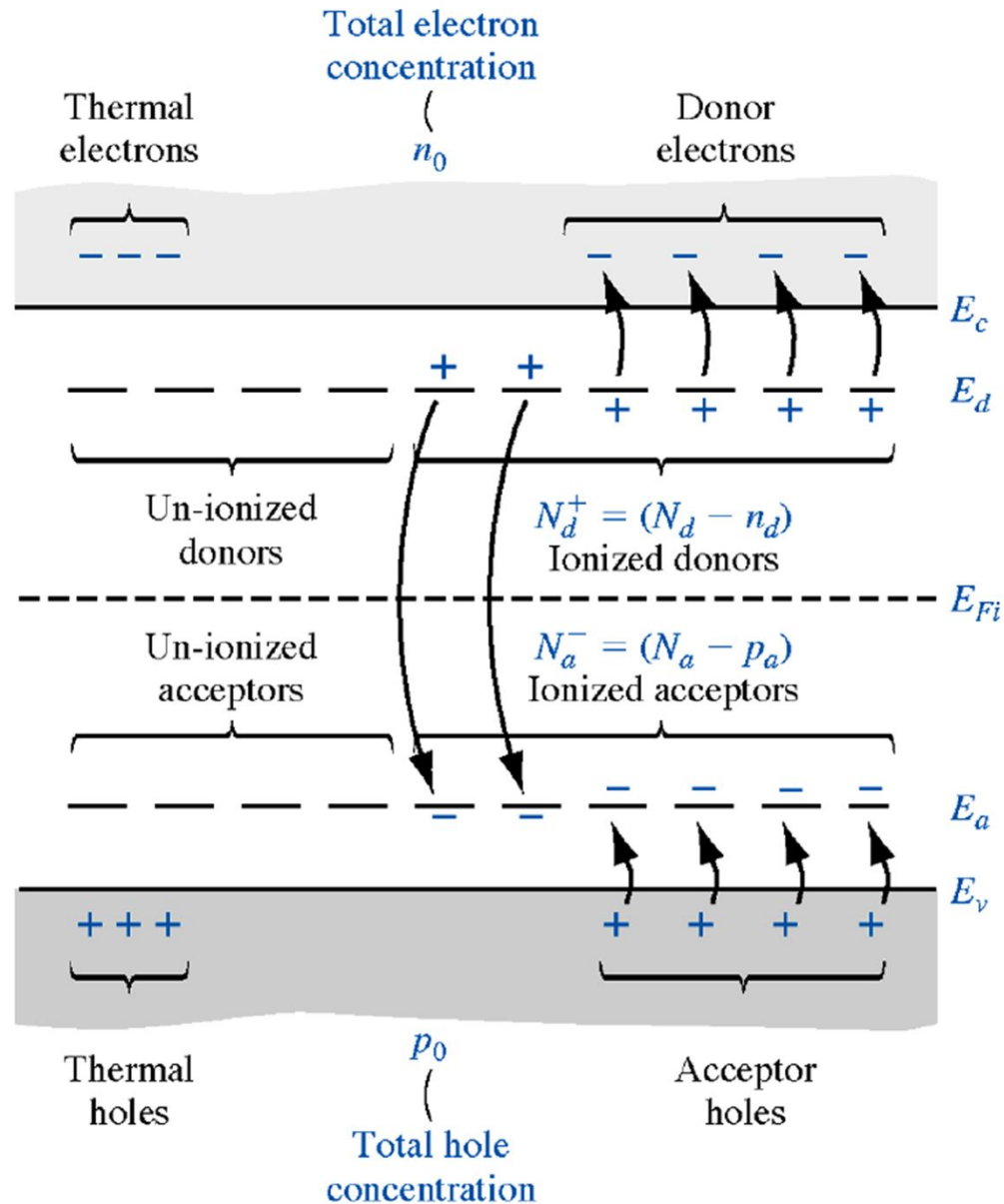
Eg. Substrate is predoped usually p-type. All other dopings are done on top of this.

4.5 | CHARGE NEUTRALITY

$$n_0 + \underbrace{(N_a - p_a)}_{\text{Ionized acceptors}} = p_0 + \underbrace{(N_d - n_d)}_{\text{Ionized donors}}$$

complete ionization
 n_d and p_a are both zero

$$n_0 + N_a = p_0 + N_d$$



Charge neutrality

Recall $n_0 + N_a = p_0 + N_d$

Using the relation $n_i^2 = n_0 p_0$

$$n_0 + N_a = \frac{n_i^2}{n_0} + N_d$$

$$n_0 = \frac{(N_d - N_a)}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$$

n_0 is not simply N_d

Charge neutrality

Similarly in p-type semiconductor,

$$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$$

Minority carrier concentration

$$p_0 = \frac{n_i^2}{n_0}$$

(n-type material)

$$n_0 = \frac{n_i^2}{p_0}$$

(p-type material)

Redistribution of electrons when donors are added

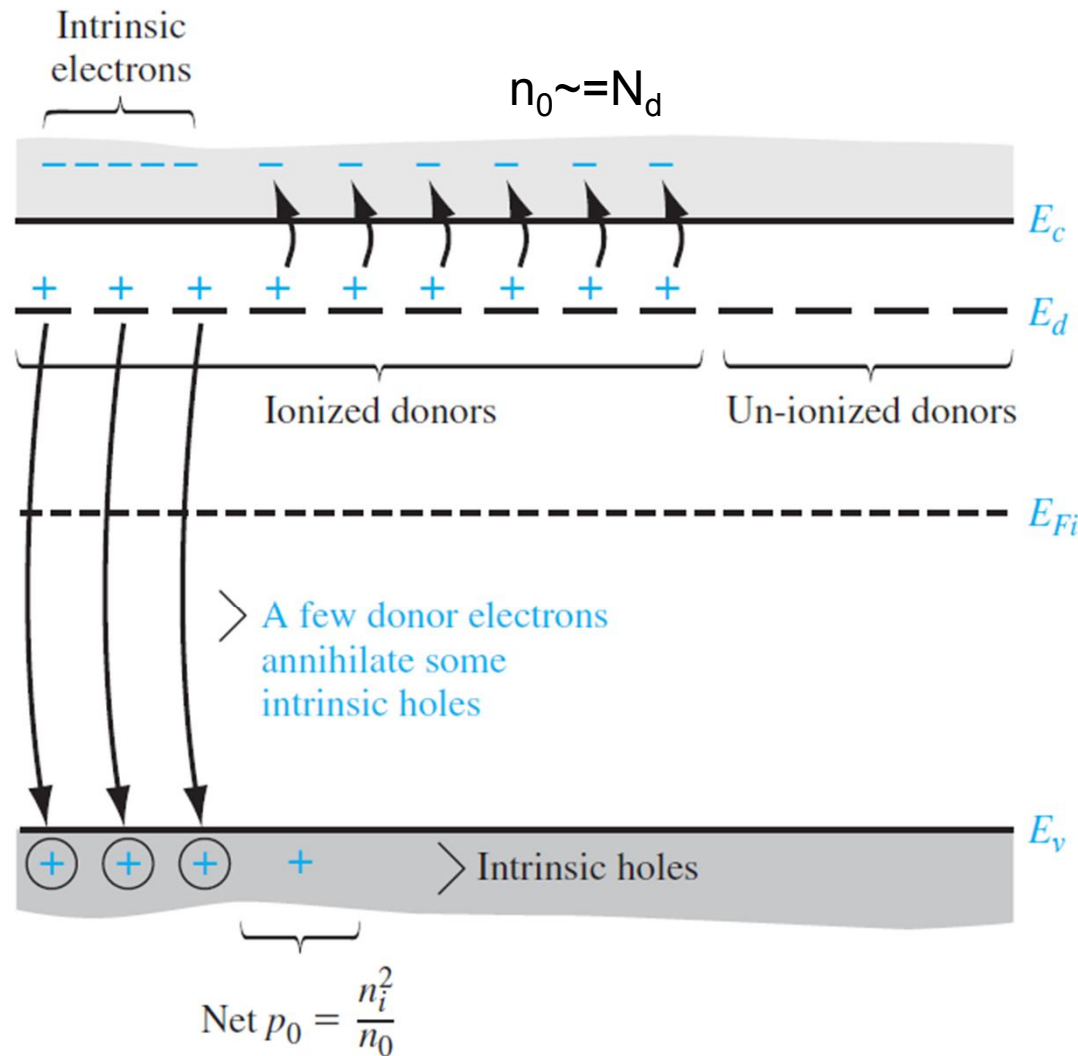


Figure 4.15 | Energy-band diagram showing the redistribution of electrons when donors are added.

4.6 | POSITION OF FERMI ENERGY LEVEL

Where is the Fermi level of an extrinsic semiconductor?

$$n_0 = N_c \exp \left[\frac{-(E_c - E_F)}{kT} \right]$$

$$E_c - E_F = kT \ln \left(\frac{N_c}{n_0} \right)$$

N-type: $N_d \gg n_i$ then $n_0 \approx N_d$

$$E_c - E_F = kT \ln \left(\frac{N_c}{N_d} \right)$$

4.6 | POSITION OF FERMI ENERGY LEVEL

Where is the Fermi level of a p-type extrinsic semiconductor?

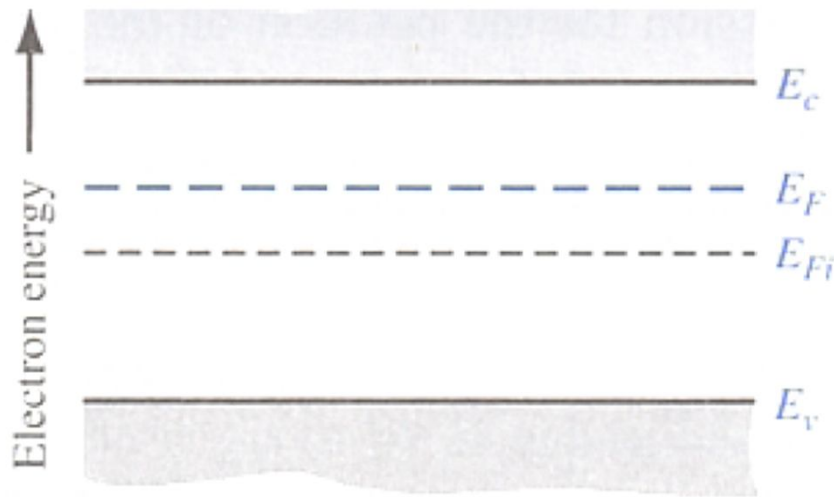
$$p_0 = N_v \exp \left[\frac{-(E_F - E_v)}{kT} \right]$$

$$E_F - E_v = kT \ln \left(\frac{N_v}{p_0} \right)$$

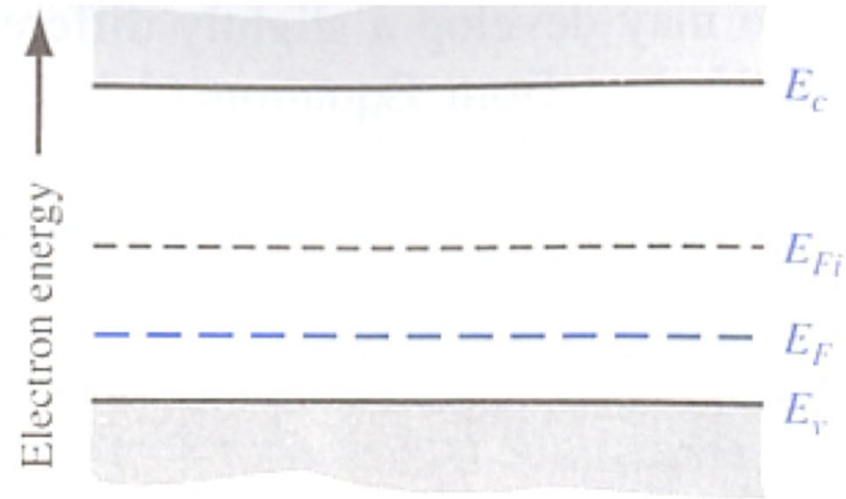
P-type: $N_a \gg n_i$ then $p_0 \approx N_a$

$$E_F - E_v = kT \ln \left(\frac{N_v}{N_a} \right)$$

4.6 | POSITION OF FERMI ENERGY LEVEL



(a)



(b)

Position of Fermi level for an (a) n-type and (b) p-type semiconductor.

4.6 | POSITION OF FERMI ENERGY LEVEL

Different expression for the n-type...

$$E_F - E_{Fi} = kT \ln \left(\frac{n_0}{n_i} \right)$$

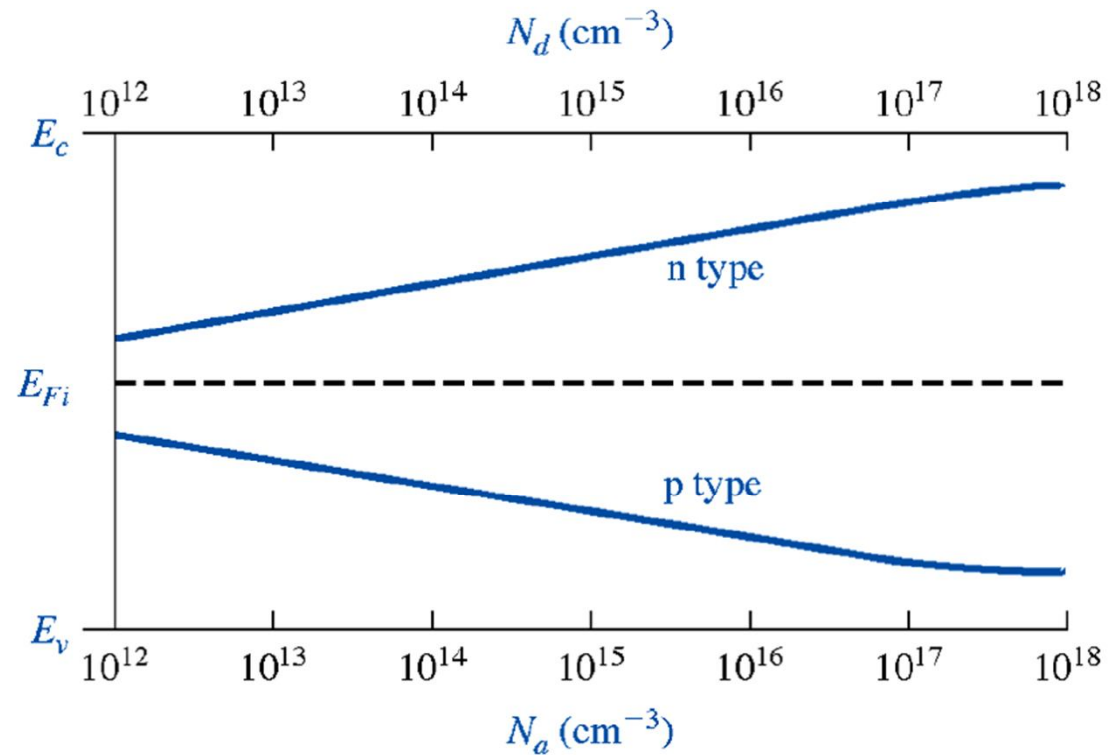
Another expression for the p-type

$$E_{Fi} - E_F = kT \ln \left(\frac{p_0}{n_i} \right)$$

Variation of E_F with doping concentration

$$E_F = E_c - kT \ln \left(\frac{N_c}{N_d} \right)$$

$$E_F = E_v + kT \ln \left(\frac{N_v}{N_a} \right)$$



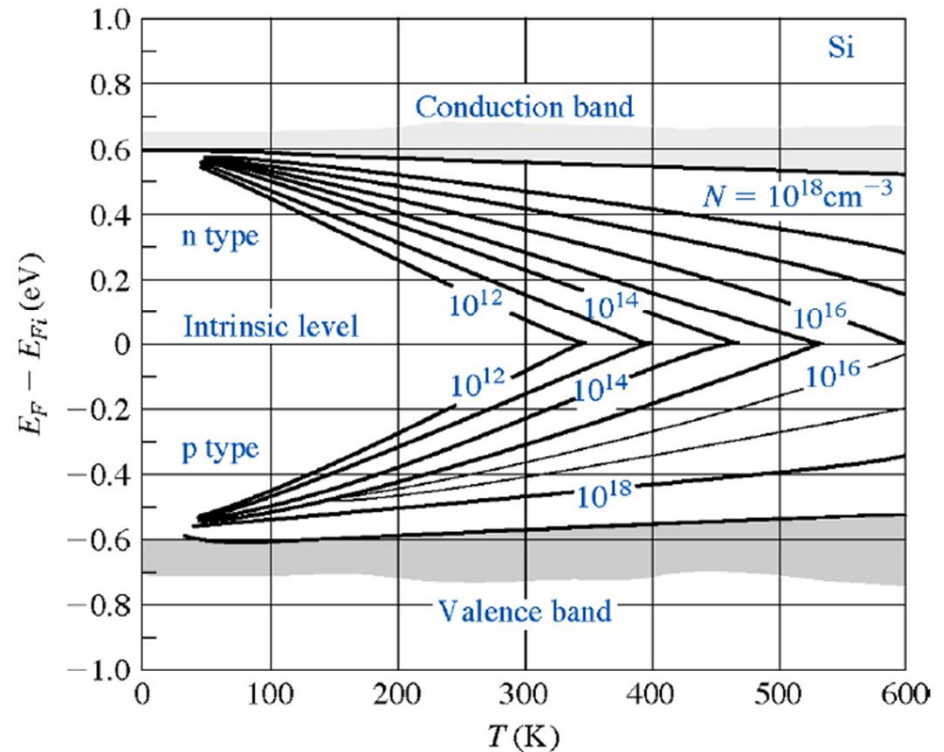
Variation of E_F with temperature T

$$E_F - E_{Fi} = kT \ln \left(\frac{n_0}{n_i} \right)$$

$$E_{Fi} - E_F = kT \ln \left(\frac{p_0}{n_i} \right)$$

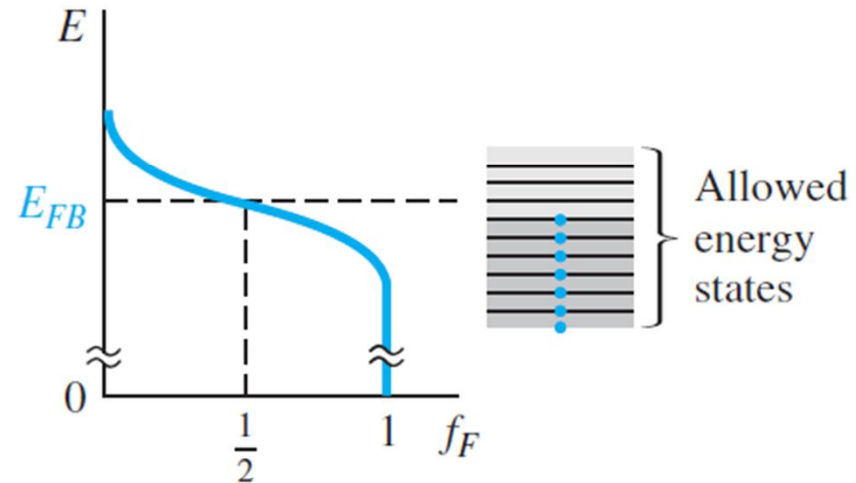
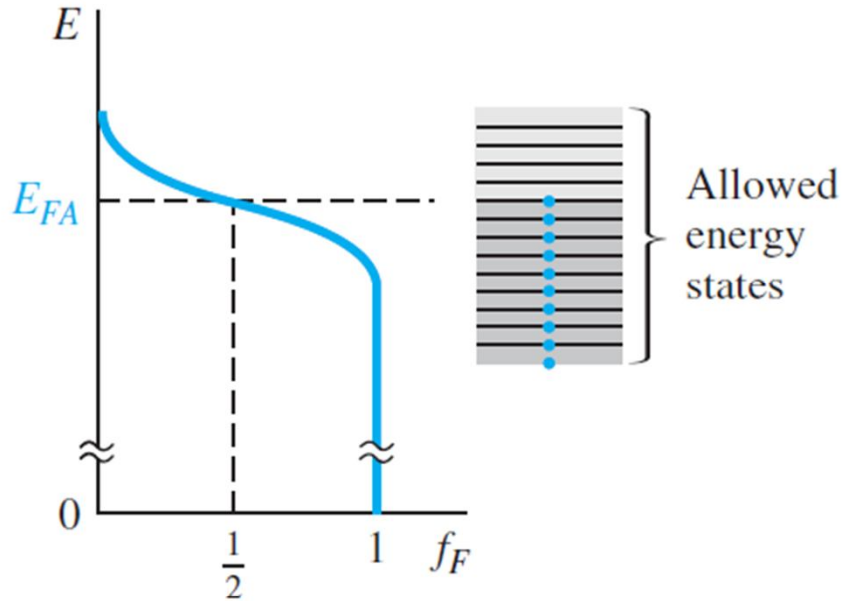
- At higher temperatures, the semiconductor becomes more intrinsic. n_i increases and Fermi level moves towards mid-gap

- At $T=0$, Fermi level is above E_d in n-type and below E_a in p-type semiconductor



Variation of Fermi level with temperature for different doping concentrations

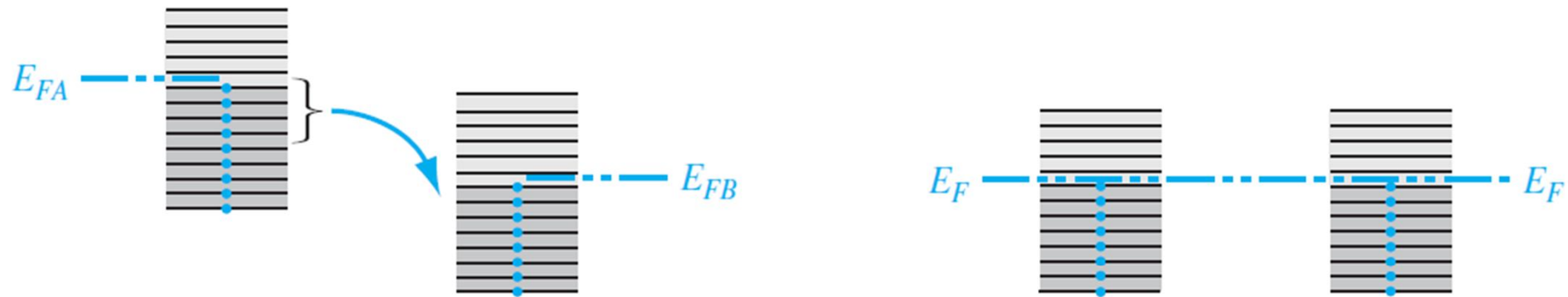
E_F must be equal when different systems are in contact and in thermodynamic equilibrium



Consider a material A, with Fermi level E_{FA} . Bands below E_{FA} are full and above are empty.

material B with Fermi level E_{FB} .

E_F must be equal when different systems are in contact and in thermodynamic equilibrium



- When A and B are brought in contact, electrons will flow from A into lower energy states of B, until thermal equilibrium is reached.
- Thermal equilibrium $\rightarrow E_F$ same in A & B

Summary

- Electron concentration $n_0 = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$
 - Hole concentration $p_0 = N_v \exp\left[\frac{-(E_F - E_v)}{kT}\right]$
- } Holds for both intrinsic as well as extrinsic semiconductor
- Intrinsic carrier concentration :

$$n_0 p_0 = n_i^2 \quad n_i^2 = N_c N_v \exp\left[\frac{-E_g}{kT}\right]$$

- In intrinsic semiconductor, Fermi level is close to but not exactly in the centre between conduction and valence bands.

$$E_{Fi} - E_{\text{midgap}} = \frac{3}{4} kT \ln \left(\frac{m_p^*}{m_n^*} \right)$$

Summary

- In extrinsic semiconductor, Fermi level is close to conduction band (n-type) or valence band (p-type)
- Position of Fermi level in extrinsic semiconductor

$$E_F - E_{Fi} = kT \ln \left(\frac{n_0}{n_i} \right)$$

- In compensated n-type semiconductor electron concentration is given by

$$n_0 = \frac{(N_d - N_a)}{2} + \sqrt{\left(\frac{N_d - N_a}{2} \right)^2 + n_i^2}$$

- When two different systems are in contact and in thermal equilibrium, E_F must be the same in both systems.

Reading Materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 4, “The Semiconductor in Equilibrium”.